TANZANIAN'S WATER WELLS PREDICTION PROJECT

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1. Overview

1.1 Business Understanding

Water Wells for Africa (WWFA) is a non-governmental organization committed to ensuring sustainable access to clean and safe water across rural and underserved communities in Africa. With a mission to bridge the water access gap, WWFA works closely with local governments, community leaders, and international partners to maintain and restore water infrastructure across the country.

1.1.1 Business Problem

Tanzania, a developing nation with a population of over 57 million people, faces a critical water access crisis. While thousands of water points have been established across the country, a significant number are either non-functional or in need of repair, leaving millions of citizens without reliable access to clean water. Currently, identifying which water pumps need attention relies heavily on manual field inspections, which are costly, time-consuming, and inefficient. By the time a faulty pump is identified and repaired, communities may have gone weeks or months without clean water, leading to serious public health consequences.

My role as a data scientist at WWFA is to analyze historical water points across Tanzania and dentify patterns of the non-functional wells that are contributing to water shortage such as pump failure,geo-location, pump type and hwo they are managed. These insights will help in coming up with recommendations to improve water wells for long-term basis.. Lastely by laveraging these patterns I will be able to build a

predictive classifiation model that will ccurately identify existing water points that are non-functional or in need of repair , ensuring that field teams can intervene quickly and efficiently, and that the people of Tanzania have reliable access to clean, potable water.

1.1.2 Objectives

1. To identify the key drivers of water point failures in Tanzania.
2. To identify the functional and non-functional water points and build a predictive classification model based on the available variables.
3. To come up with actionable insights for the future water point constructions in Tanzania.

1.1.3 Reasearch Questions

1. What are the key drivers of water point failures in Tanzania?
2. Which water points are functional and non-functional, and can a predictive classification model be built based on the available variables?
3. What actionable insights can be drawn to guide future water point constructions in Tanzania?

1.1.4 Success Metrics

1. Successfully identify and rank the top features contributing to water point failure through feature importance analysis and exploratory data analysis.
2. Successfully build a Predictive Classification Model that will be able to predict and improve water well stituation in Tanzania with an accuracy of at least 80%.
3. To successfully come up with guides for future water point constructions in Tanzania

1.2 Data Understanding

Data Understanding helps us know where the data came from and how they relate to the data anaysis questions It helps us know what and who is in the data presented, and the properties of the variables to be used

1.2.1 Importing necessary libraries:

In [1]:
```
import numpy as np
import pandas as pd
pd.set_option("display.max_columns", None)
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
import warnings
warnings.filterwarnings("ignore")

from sklearn.svm import SVC
from sklearn.model_selection import ShuffleSplit, GridSearchCV
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression, Lasso, LinearRegression
from sklearn.naive_bayes import GaussianNB, MultinomialNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import StandardScaler
from sklearn import tree
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (classification_report, confusion_matrix,
                             ConfusionMatrixDisplay, accuracy_score,
                             precision_score, recall_score, f1_score)
```

1.2.2 Loading and Merging the 3 Dataset

In [2]:
```
# Load the data into pandas DataFrame
train = pd.read_csv('water_well_1.csv')
labels = pd.read_csv('train_label_1.csv')
test = pd.read_csv('water_well_2.csv')
```

```
# Merge to get full training dataset
df = train.merge(labels, on='id')
print(df.shape)
```

```
(59400, 41)
```

The dataset consists of 41 columns and 59400 rows.

1.2.3 Peviewing the Data

In [3]:
```
# Preview the DataFrame
df.head(5)
```

Out[3]:

	id	amount_tsh	date_recorded	funder	gps_height	installer	longitude	latitude	wpt_name	num_private	basin	subvilla
0	69572	6000.0	2011-03-14	Roman	1390	Roman	34.938093	-9.856322	none	0	Lake Nyasa	Mnyus
1	8776	0.0	2013-03-06	Grumeti	1399	GRUMETI	34.698766	-2.147466	Zahanati	0	Lake Victoria	Nyama
2	34310	25.0	2013-02-25	Lottery Club	686	World vision	37.460664	-3.821329	Kwa Mahundi	0	Pangani	Majen
3	67743	0.0	2013-01-28	Unicef	263	UNICEF	38.486161	-11.155298	Zahanati Ya Nanyumbu	0	Ruvuma / Southern Coast	Mahakama
4	19728	0.0	2011-07-13	Action In A	0	Artisan	31.130847	-1.825359	Shuleni	0	Lake Victoria	Kyanyam

In [4]:
```
# Describe the DataFrame
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 59400 entries, 0 to 59399
Data columns (total 41 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   id                     59400 non-null  int64
 1   amount_tsh             59400 non-null  float64
 2   date_recorded          59400 non-null  object
 3   funder                 55763 non-null  object
 4   gps_height             59400 non-null  int64
 5   installer              55745 non-null  object
 6   longitude              59400 non-null  float64
 7   latitude               59400 non-null  float64
 8   wpt_name               59398 non-null  object
 9   num_private            59400 non-null  int64
 10  basin                  59400 non-null  object
 11  subvillage             59029 non-null  object
 12  region                 59400 non-null  object
 13  region_code            59400 non-null  int64
 14  district_code          59400 non-null  int64
 15  lga                    59400 non-null  object
 16  ward                   59400 non-null  object
 17  population             59400 non-null  int64
 18  public_meeting         56066 non-null  object
 19  recorded_by            59400 non-null  object
 20  scheme_management      55522 non-null  object
 21  scheme_name            30590 non-null  object
 22  permit                 56344 non-null  object
 23  construction_year      59400 non-null  int64
 24  extraction_type        59400 non-null  object
 25  extraction_type_group  59400 non-null  object
 26  extraction_type_class  59400 non-null  object
 27  management             59400 non-null  object
 28  management_group       59400 non-null  object
 29  payment                59400 non-null  object
 30  payment_type           59400 non-null  object
 31  water_quality          59400 non-null  object
 32  quality_group          59400 non-null  object
 33  quantity               59400 non-null  object
 34  quantity_group         59400 non-null  object
 35  source                 59400 non-null  object
```

```
 36  source_type            59400 non-null  object
 37  source_class           59400 non-null  object
 38  waterpoint_type        59400 non-null  object
 39  waterpoint_type_group  59400 non-null  object
 40  status_group           59400 non-null  object
dtypes: float64(3), int64(7), object(31)
memory usage: 18.6+ MB
```

In [5]:

```python
# Checking the numerical and categorical columns

# Numerical Columns
print(f"Numerical Columns: {df.select_dtypes(include='number').columns.tolist()}\n")

# Categorical Columns
print(f"Categorical Columns: {df.select_dtypes(include='object').columns.tolist()}")
```

```
Numerical Columns: ['id', 'amount_tsh', 'gps_height', 'longitude', 'latitude', 'num_private', 'region_code', 'district_code',
'population', 'construction_year']

Categorical Columns: ['date_recorded', 'funder', 'installer', 'wpt_name', 'basin', 'subvillage', 'region', 'lga', 'ward', 'publ
ic_meeting', 'recorded_by', 'scheme_management', 'scheme_name', 'permit', 'extraction_type', 'extraction_type_group', 'extracti
on_type_class', 'management', 'management_group', 'payment', 'payment_type', 'water_quality', 'quality_group', 'quantity', 'qua
ntity_group', 'source', 'source_type', 'source_class', 'waterpoint_type', 'waterpoint_type_group', 'status_group']
```

In [6]:

```python
# Describe the  data
df.describe()
```

Out[6]:

	id	amount_tsh	gps_height	longitude	latitude	num_private	region_code	district_code	population
count	59400.000000	59400.000000	59400.000000	59400.000000	5.940000e+04	59400.000000	59400.000000	59400.000000	59400.000000
mean	37115.131768	317.650385	668.297239	34.077427	-5.706033e+00	0.474141	15.297003	5.629747	179.909983
std	21453.128371	2997.574558	693.116350	6.567432	2.946019e+00	12.236230	17.587406	9.633649	471.482176
min	0.000000	0.000000	-90.000000	0.000000	-1.164944e+01	0.000000	1.000000	0.000000	0.000000
25%	18519.750000	0.000000	0.000000	33.090347	-8.540621e+00	0.000000	5.000000	2.000000	0.000000
50%	37061.500000	0.000000	369.000000	34.908743	-5.021597e+00	0.000000	12.000000	3.000000	25.000000
75%	55656.500000	20.000000	1319.250000	37.178387	-3.326156e+00	0.000000	17.000000	5.000000	215.000000
max	74247.000000	350000.000000	2770.000000	40.345193	-2.000000e-08	1776.000000	99.000000	80.000000	30500.000000

1.3 Data Preparation

Data preparation purpose is to clean the data, select necessary columns, and conduct Exploratory Data Anaysis in order to have clean data for modelling and accuracy

In [7]:

```
#preview data again
df
```

Out[7]:

	id	amount_tsh	date_recorded	funder	gps_height	installer	longitude	latitude	wpt_name	num_private	basin	s
0	69572	6000.0	2011-03-14	Roman	1390	Roman	34.938093	-9.856322	none	0	Lake Nyasa	
1	8776	0.0	2013-03-06	Grumeti	1399	GRUMETI	34.698766	-2.147466	Zahanati	0	Lake Victoria	
2	34310	25.0	2013-02-25	Lottery Club	686	World vision	37.460664	-3.821329	Kwa Mahundi	0	Pangani	
3	67743	0.0	2013-01-28	Unicef	263	UNICEF	38.486161	-11.155298	Zahanati Ya Nanyumbu	0	Ruvuma / Southern Coast	Ma
4	19728	0.0	2011-07-13	Action In A	0	Artisan	31.130847	-1.825359	Shuleni	0	Lake Victoria	Ky
...	...	...	...	...	...	...	...	...	...	...	...	
59395	60739	10.0	2013-05-03	Germany Republi	1210	CES	37.169807	-3.253847	Area Three Namba 27	0	Pangani	
59396	27263	4700.0	2011-05-07	Cefa-njombe	1212	Cefa	35.249991	-9.070629	Kwa Yahona Kuvala	0	Rufiji	
59397	37057	0.0	2011-04-11	NaN	0	NaN	34.017087	-8.750434	Mashine	0	Rufiji	M
59398	31282	0.0	2011-03-08	Malec	0	Musa	35.861315	-6.378573	Mshoro	0	Rufiji	

	id	amount_tsh	date_recorded	funder	gps_height	installer	longitude	latitude	wpt_name	num_private	basin	s
59399	26348	0.0	2011-03-23	World Bank	191	World	38.104048	-6.747464	Kwa Mzee Lugawa	0	Wami / Ruvu	Kika

59400 rows × 41 columns

1.3.1 Checking related columns

Since many of the 41 columns in the dataset appear to be interrelated, I will start by examining the correlated variables before proceeding to analyze the remaining columns.

3.1.1.1 Related column 1(scheme management, management, management_group)

In [8]:
```
# checking unique values in scheme_management column
df['scheme_management'].value_counts(dropna=False)
```

Out[8]:
```
scheme_management
VWC                 36793
WUG                  5206
NaN                  3878
Water authority      3153
WUA                  2883
Water Board          2748
Parastatal           1680
Private operator     1063
Company              1061
Other                 766
SWC                    97
Trust                  72
Name: count, dtype: int64
```

In [9]:
```
# checking unique values in management column
df['management'].value_counts(dropna=False)
```

Out[9]:
```
management
vwc                 40507
wug                  6515
water board          2933
wua                  2535
private operator     1971
parastatal           1768
water authority       904
other                 844
company               685
unknown               561
other - school         99
trust                  78
Name: count, dtype: int64
```

In [10]:
```
# checking unique values in management_group column
df['management_group'].value_counts(dropna=False)
```

Out[10]:
```
management_group
user-group    52490
commercial     3638
parastatal     1768
other           943
unknown         561
Name: count, dtype: int64
```

from the preview on checking related columns scheme_management and management seems to be related as for management_group appears to have fewer unique values to the 2.. In order to understand how the two now relate I will perform a group by to examint their relationship

In [11]:
```
# Group by management_group only since management column has already been dropped in next cell to avoid errors
summary = (
    df.groupby(['management_group'])
    .agg(
        count=('management_group', 'size'),
        percentage=('management_group', lambda x: round((len(x) / len(df)) * 100, 2))
    )
    .rename_axis(['Management Group'])
    .sort_values(by='count', ascending=False)
```

```
)

# Add a cumulative percentage column
summary['cumulative_%'] = summary['percentage'].cumsum().round(2)

# Display the result
summary.style.background_gradient(cmap='Blues', subset=['count', 'percentage'])
```

Out[11]:

Management Group	count	percentage	cumulative_%
user-group	52490	88.370000	88.370000
commercial	3638	6.120000	94.490000
parastatal	1768	2.980000	97.470000
other	943	1.590000	99.060000
unknown	561	0.940000	100.000000

From the output, I can observe a clear relationship between the management_group and management columns. However, since management_group contains fewer unique values than management, I will drop the management column to avoid model complexity.

In [12]:

```
# dropping the management column
# Drop the management and scheme_management columns if they exist
df = df.drop(['scheme_management', 'management'], axis=1, errors='ignore')

# Confirm the columns have been dropped
print(df.columns.tolist())
```

```
['id', 'amount_tsh', 'date_recorded', 'funder', 'gps_height', 'installer', 'longitude', 'latitude', 'wpt_name', 'num_private',
'basin', 'subvillage', 'region', 'region_code', 'district_code', 'lga', 'ward', 'population', 'public_meeting', 'recorded_by',
'scheme_name', 'permit', 'construction_year', 'extraction_type', 'extraction_type_group', 'extraction_type_class', 'management_
group', 'payment', 'payment_type', 'water_quality', 'quality_group', 'quantity', 'quantity_group', 'source', 'source_type', 'so
urce_class', 'waterpoint_type', 'waterpoint_type_group', 'status_group']
```

1.3.1.2 Related column 2(payment, payment type)

In [13]:
```
# checking unique values in payment column
df['payment'].value_counts(dropna=False)
```

Out[13]:
```
payment
never pay                25348
pay per bucket            8985
pay monthly               8300
unknown                   8157
pay when scheme fails     3914
pay annually              3642
other                     1054
Name: count, dtype: int64
```

In [14]:
```
# checking unique values in the payment_type column
df['payment_type'].value_counts(dropna=False)
```

Out[14]:
```
payment_type
never pay     25348
per bucket     8985
monthly        8300
unknown        8157
on failure     3914
annually       3642
other          1054
Name: count, dtype: int64
```

the two are not much related I will drop payment column

In [15]:
```
# dropping the Payment column
df = df.drop(['payment'], axis=1, errors='ignore')
```

1.3.1.3 Related column 2(extraction_type_class,extraction_type_group,extraction_type)

In [16]:
```
# checking unique values in extraction_type_class column
df['extraction_type_class'].value_counts(dropna=False)
```

Out[16]:
```
extraction_type_class
gravity         26780
handpump        16456
other            6430
submersible      6179
motorpump        2987
rope pump         451
wind-powered      117
Name: count, dtype: int64
```

In [17]:
```
# checking unique values in extraction_type_group column
df['extraction_type_group'].value_counts(dropna=False)
```

Out[17]:
```
extraction_type_group
gravity            26780
nira/tanira         8154
other               6430
submersible         6179
swn 80              3670
mono                2865
india mark ii       2400
afridev             1770
rope pump            451
other handpump       364
other motorpump      122
wind-powered         117
india mark iii        98
Name: count, dtype: int64
```

In [18]:
```
# checking unique values in extraction_type column
df['extraction_type'].value_counts(dropna=False)
```

Out[18]:
```
extraction_type
gravity                      26780
nira/tanira                   8154
other                         6430
submersible                   4764
swn 80                        3670
mono                          2865
india mark ii                 2400
afridev                       1770
ksb                           1415
other - rope pump              451
other - swn 81                 229
windmill                       117
india mark iii                  98
cemo                            90
other - play pump               85
walimi                          48
climax                          32
other - mkulima/shinyanga        2
Name: count, dtype: int64
```

extraction_type and extraction_type_group are related while extraction_type_class is not related to the two.

In [19]:
```
# Group by extraction_type_class, extraction_type_group, and extraction_type with aggregated statistics
summary = (
    df.groupby(['extraction_type_class', 'extraction_type_group', 'extraction_type'])
    .agg(
        count=('extraction_type', 'size'),
        percentage=('extraction_type', lambda x: round((len(x) / len(df)) * 100, 2))
    )
    .rename_axis(['Extraction Type Class', 'Extraction Type Group', 'Extraction Type'])
    .sort_values(by='count', ascending=False)
)

# Add a cumulative percentage column
summary['cumulative_%'] = summary['percentage'].cumsum().round(2)

# Display the result with styling
summary.style.background_gradient(cmap='Blues', subset=['count', 'percentage', 'cumulative_%'])
```

Out[19]:

Extraction Type Class	Extraction Type Group	Extraction Type	count	percentage	cumulative_%
gravity	gravity	gravity	26780	45.080000	45.080000
handpump	nira/tanira	nira/tanira	8154	13.730000	58.810000
other	other	other	6430	10.820000	69.630000
submersible	submersible	submersible	4764	8.020000	77.650000
handpump	swn 80	swn 80	3670	6.180000	83.830000
motorpump	mono	mono	2865	4.820000	88.650000
handpump	india mark ii	india mark ii	2400	4.040000	92.690000
	afridev	afridev	1770	2.980000	95.670000
submersible	submersible	ksb	1415	2.380000	98.050000
rope pump	rope pump	other - rope pump	451	0.760000	98.810000
handpump	other handpump	other - swn 81	229	0.390000	99.200000
wind-powered	wind-powered	windmill	117	0.200000	99.400000
handpump	india mark iii	india mark iii	98	0.160000	99.560000
motorpump	other motorpump	cemo	90	0.150000	99.710000
handpump	other handpump	other - play pump	85	0.140000	99.850000
		walimi	48	0.080000	99.930000
motorpump	other motorpump	climax	32	0.050000	99.980000
handpump	other handpump	other - mkulima/shinyanga	2	0.000000	99.980000

Extraction_type_group seems nore detalied, I will drop extraction_type_class and extraction_type clolumns

In [20]:
```
# Droping the extraction_type_class and extraction_type columns
```

```
df=df.drop(['extraction_type_class', 'extraction_type'], axis=1, errors='ignore')
```

3.1.1.4 Related column 4(Quality and Quantity group)

In [21]:
```
# checking unique values in Quality column
df['quantity'].value_counts(dropna=False)
```

Out[21]:
```
quantity
enough          33186
insufficient    15129
dry              6246
seasonal         4050
unknown           789
Name: count, dtype: int64
```

In [22]:
```
# checking unique values in Quality column
df['quantity_group'].value_counts(dropna=False)
```

Out[22]:
```
quantity_group
enough          33186
insufficient    15129
dry              6246
seasonal         4050
unknown           789
Name: count, dtype: int64
```

The two are perfectly correlated will proceed to drop one,'quantity group'

In [23]:
```
# Dropping the quantity_group column

df = df.drop(['quantity_group'], axis=1, errors='ignore')
```

3.1.1.5 Related column 5(waterpoint_type, waterpoint_type_group)

In [24]:
```
# checking unique values in waterpoint_type column
df['waterpoint_type'].value_counts(dropna=False)
```

Out[24]:
```
waterpoint_type
communal standpipe             28522
hand pump                      17488
other                           6380
communal standpipe multiple     6103
improved spring                  784
cattle trough                    116
dam                                7
Name: count, dtype: int64
```

In [25]:
```
# checking unique values in waterpoint_type_group column
df['waterpoint_type_group'].value_counts(dropna=False)
```

Out[25]:
```
waterpoint_type_group
communal standpipe    34625
hand pump             17488
other                  6380
improved spring         784
cattle trough           116
dam                       7
Name: count, dtype: int64
```

dropping waterpoint_type_group column has less unique values

In [26]:
```
# Drop the waterpoint_type_group column
df = df.drop(['waterpoint_type_group'], axis=1, errors='ignore')
```

3.1.1.6 Related column 6(source_class,source,source_type)

In [27]:
```
# checking unique values in source column
df['source'].value_counts(dropna=False)
```

Out[27]:
```
source
spring                  17021
shallow well            16824
machine dbh             11075
river                    9612
rainwater harvesting     2295
hand dtw                  874
lake                      765
dam                       656
other                     212
unknown                    66
Name: count, dtype: int64
```

In [28]:
```
# checking unique values in source_class column
df['source_class'].value_counts(dropna=False)
```

Out[28]:
```
source_class
groundwater    45794
surface        13328
unknown          278
Name: count, dtype: int64
```

In [29]:
```
# checking unique values in source_type column
df['source_type'].value_counts(dropna=False)
```

Out[29]:
```
source_type
spring                  17021
shallow well            16824
borehole                11949
river/lake              10377
rainwater harvesting     2295
dam                       656
other                     278
Name: count, dtype: int64
```

dropping source,source_class column has less unique values

In [30]:
```
# Drop the 'source' and 'source_class' columns
df = df.drop(['source', 'source_class'], axis=1, errors='ignore')
```

1.3.1.5 Dropping all other columns that are not useful

drop all columns that will not help in coming up with the perfect predictive model

In [31]:
```
# Drop all unnecessary columns
df = df.drop([
    'longitude', 'latitude', 'wpt_name', 'num_private',
    'region_code', 'district_code', 'public_meeting',
    'recorded_by', 'scheme_name'
], axis=1, errors='ignore')

# Confirm the columns have been dropped
print(df.columns.tolist())
```

```
['id', 'amount_tsh', 'date_recorded', 'funder', 'gps_height', 'installer', 'basin', 'subvillage', 'region', 'lga', 'ward', 'pop
ulation', 'permit', 'construction_year', 'extraction_type_group', 'management_group', 'payment_type', 'water_quality', 'quality
_group', 'quantity', 'source_type', 'waterpoint_type', 'status_group']
```

1.4 Cleaning Data

1.4.1.checking missing values and duplicates

In [32]:
```
# Check for missing values
print("Missing Values:")
print(df.isnull().sum())
print(f"\nTotal Missing Values: {df.isnull().sum().sum()}")

print("\n" + "="*50)

# Check for duplicates
print(f"\nTotal Duplicate Rows: {df.duplicated().sum()}")
```

```
Missing Values:
id                         0
amount_tsh                 0
date_recorded              0
funder                  3637
gps_height                 0
installer               3655
basin                      0
subvillage               371
region                     0
lga                        0
ward                       0
population                 0
permit                  3056
construction_year          0
extraction_type_group      0
management_group           0
payment_type               0
water_quality              0
quality_group              0
quantity                   0
source_type                0
waterpoint_type            0
status_group               0
dtype: int64

Total Missing Values: 10719

==================================================

Total Duplicate Rows: 0
```

In [33]:

```python
# 1. Drop duplicate rows
df = df.drop_duplicates()
print(f"Shape after dropping duplicates: {df.shape}")

# 2. Fill missing values for categorical columns with mode
# funder, installer, subvillage, permit have missing values
cat_cols_with_missing = ['funder', 'installer', 'subvillage', 'permit']
for col in cat_cols_with_missing:
    df[col].fillna(df[col].mode()[0], inplace=True)
```

```
# 3. Confirm no missing values remain
print(f"\nRemaining Missing Values: {df.isnull().sum().sum()}")
print(f"\nFinal Shape: {df.shape}")
```

```
Shape after dropping duplicates: (59400, 23)

Remaining Missing Values: 0

Final Shape: (59400, 23)
```

data is clean nomissing values

In [34]:
```
# Find the number of duplicated records
print(f"Total Duplicate Rows: {df.duplicated().sum()}")

# Display the duplicated records
df[df.duplicated()]
```

```
Total Duplicate Rows: 0
```

Out[34]:

id	amount_tsh	date_recorded	funder	gps_height	installer	basin	subvillage	region	lga	ward	population	permit	construction_yea

data does not contain duplicates

In [35]:
```
# feature engineering
# Extract year and month from date_recorded
df['year_recorded'] = pd.to_datetime(df['date_recorded']).dt.year
df['month_recorded'] = pd.to_datetime(df['date_recorded']).dt.month

# Create age of water point feature
df['waterpoint_age'] = df['year_recorded'] - df['construction_year']

# Drop the date_recorded column as it is no longer needed
df = df.drop(columns=['date_recorded'], errors='ignore')

# Confirm the new features
print(df[['year_recorded', 'month_recorded', 'waterpoint_age']].head())
```

```
   year_recorded  month_recorded  waterpoint_age
0           2011               3              12
1           2013               3               3
2           2013               2               4
3           2013               1              27
4           2011               7            2011
```

1.4.2.checking outliers

Outliers are data points that deviate significantly from the rest of the dataset. They can arise from various sources, including:

- **Measurement error** — inaccuracies from instruments or tools used during data collection
- **Data entry error** — mistakes made by humans when recording data
- **Changes in system behaviour** — unexpected shifts in how a system operates
- **Fraudulent behaviour** — manipulated or falsified data points
- **Natural deviations** — genuine variations that occur naturally within a population

Columns for Outlier Treatment

I will perform outlier treatment on the following numerical columns in the dataset:

- `amount_tsh`
- `gps_height`
- `population`
- `construction_year`

Why Outlier Treatment is Necessary

Outlier treatment is necessary because outliers can:

- Increase error variance and reduce the power of statistical tests
- Decrease normality if they are non-randomly distributed

Methods for Detecting Outliers

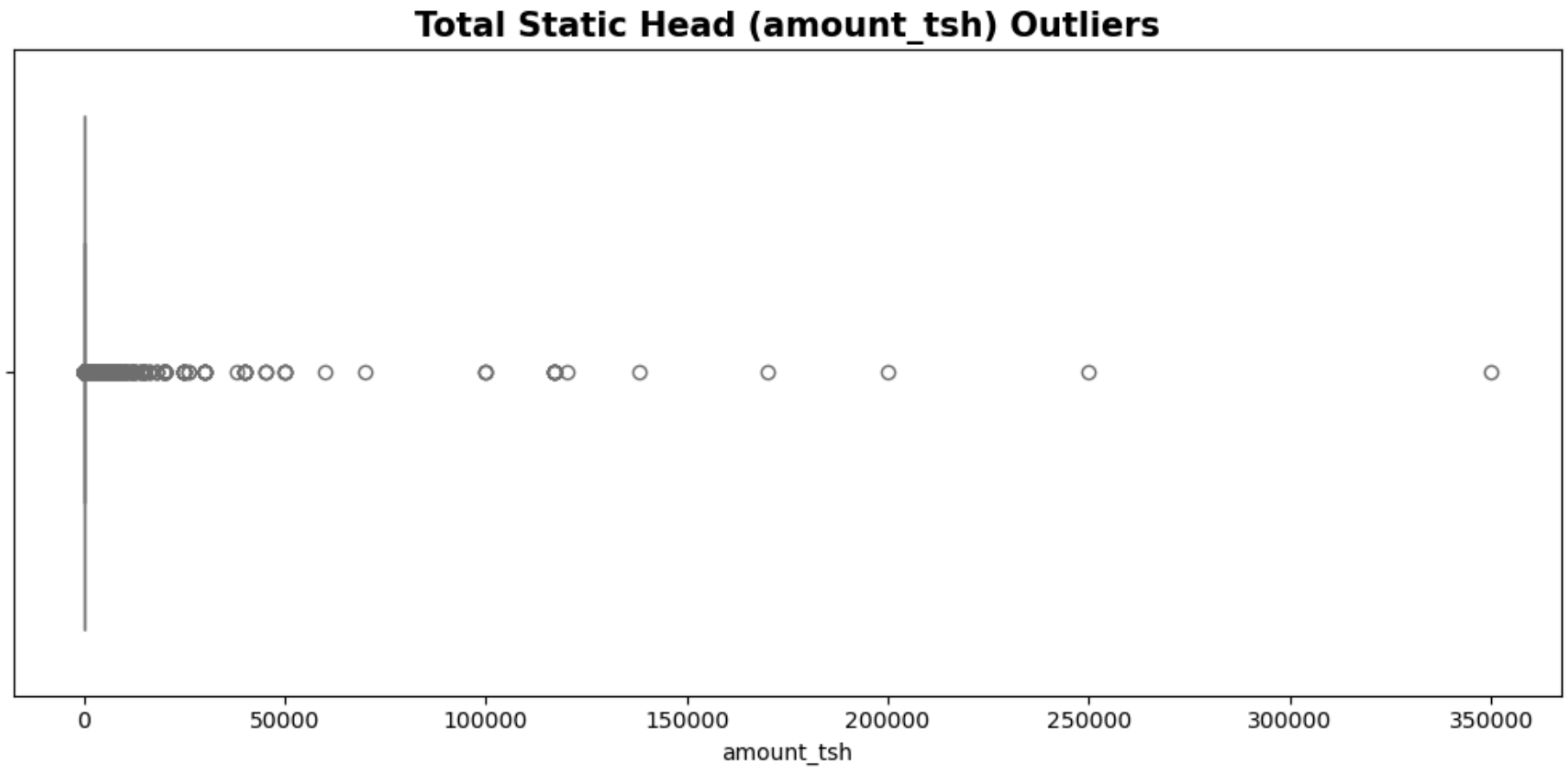
To detect outliers, I will use the following methods:

- **Boxplots** — visually displays the distribution of data and highlights outliers

as points beyond the whiskers

In [36]:
```
def outliers(df, col, title):
    """
    This function visualizes outliers using a boxplot.

    Parameters:
    -----------
    df  : pandas DataFrame
    col : column to visualize
    title : title of the plot
    """
    # Calculate outlier boundaries
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outlier_count = df[(df[col] < lower_bound) | (df[col] > upper_bound)].shape[0]

    # Create a single boxplot
    plt.figure(figsize=(10, 4))
    sns.boxplot(x=df[col], color='skyblue')
    plt.title(title, fontsize=15, fontweight='bold')
    plt.tight_layout()
    plt.show()

    # Print outlier summary
    print(f"Lower Bound        : {lower_bound:.2f}")
    print(f"Upper Bound        : {upper_bound:.2f}")
    print(f"Number of Outliers : {outlier_count}")
```

In [37]:
```
# Visualise the outliers in the amount_tsh column
plt.figure(figsize=(10, 5))
sns.boxplot(x=df['amount_tsh'], color='skyblue')
plt.title('Total Static Head (amount_tsh) Outliers', fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()
```



Conclusion: amount_tsh Outliers

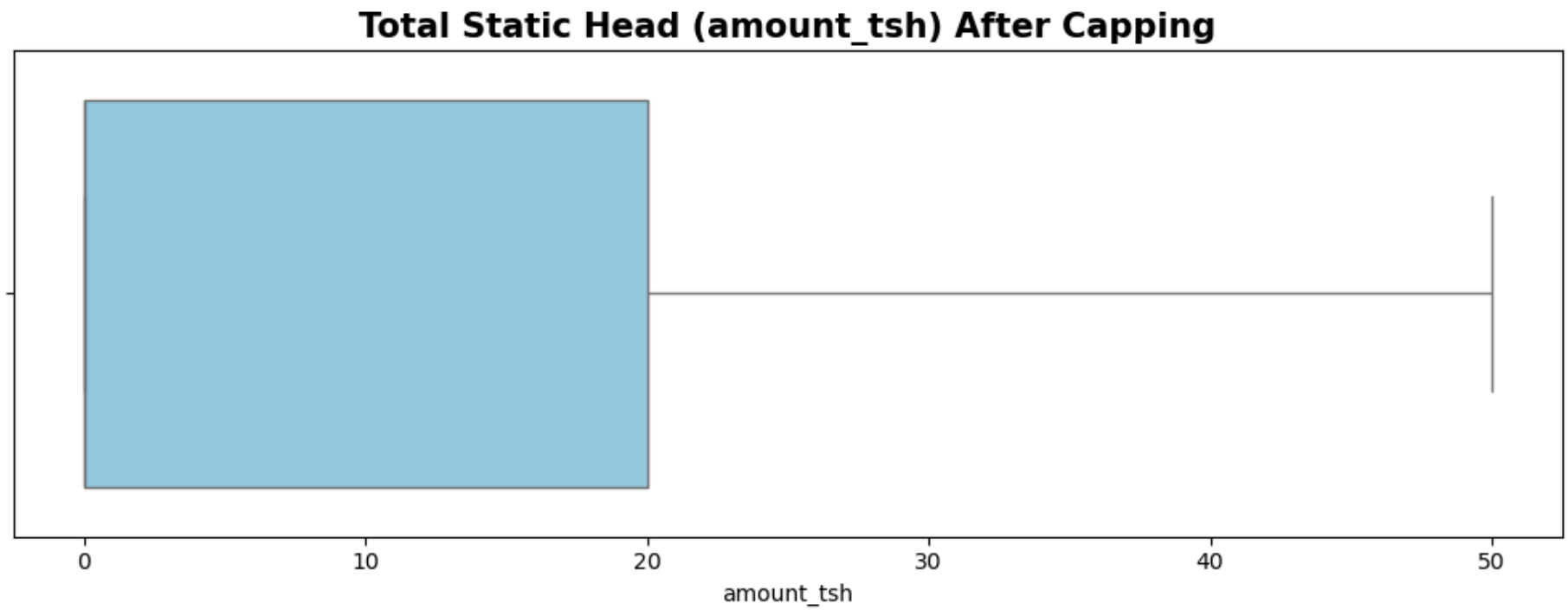
From the boxplot, we can observe that the `amount_tsh` column contains extreme outliers, with values reaching up to 350,000. The majority of the data is concentrated near 0, indicating that the distribution is heavily right-skewed.

To treat these outliers, we will use the capping (Winsorization) method, which replaces the outliers with the upper bound value. This method is preferred because it retains all rows in the dataset while reducing the effect of extreme values.

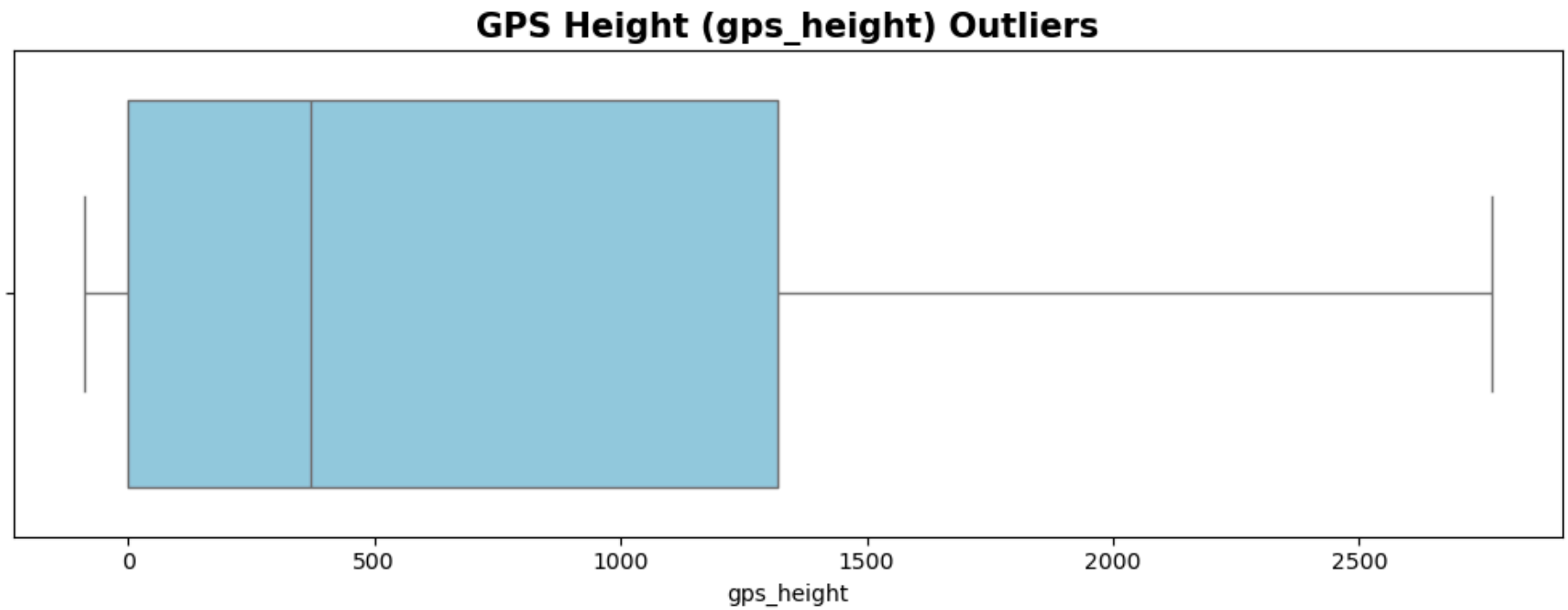
In [38]:
```
# Calculate the upper bound for amount_tsh
Q1 = df['amount_tsh'].quantile(0.25)
Q3 = df['amount_tsh'].quantile(0.75)
IQR = Q3 - Q1
upper_bound = Q3 + 1.5 * IQR

print(f"Upper Bound: {upper_bound}")

# Cap the outliers at the upper bound
df['amount_tsh'] = df['amount_tsh'].clip(upper=upper_bound)

# Confirm the capping worked
print(f"Max value after capping: {df['amount_tsh'].max()}")
```

```
Upper Bound: 50.0
Max value after capping: 50.0
```

In [39]:
```
# Visualise the amount_tsh column after capping
plt.figure(figsize=(10, 4))
sns.boxplot(x=df['amount_tsh'], color='skyblue')
plt.title('Total Static Head (amount_tsh) After Capping', fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()
```



In [40]:
```
# Visualise the outliers in the gps_height column
outliers(df, 'gps_height', 'GPS Height (gps_height) Outliers')
```

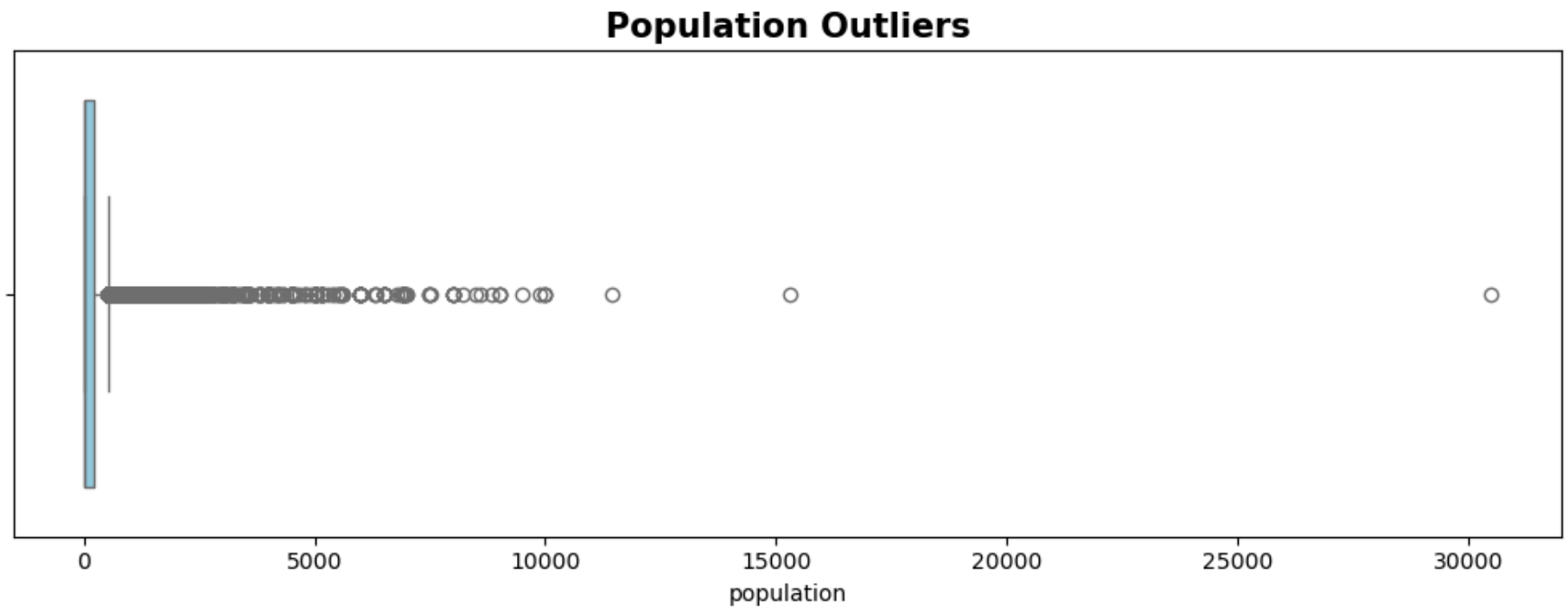



```
Lower Bound        : -1978.88
Upper Bound        : 3298.12
Number of Outliers : 0
```

Conclusion: gps_height Outliers

From the boxplot, we can observe that the gps_height column contains no outliers. Therefore, no outlier treatment is required for this column.

In [41]:
```
# Visualise the outliers in the population column
outliers(df, 'population', 'Population Outliers')
```




```
Lower Bound        : -322.50
Upper Bound        : 537.50
Number of Outliers : 4383
```

Conclusion: Population Outliers

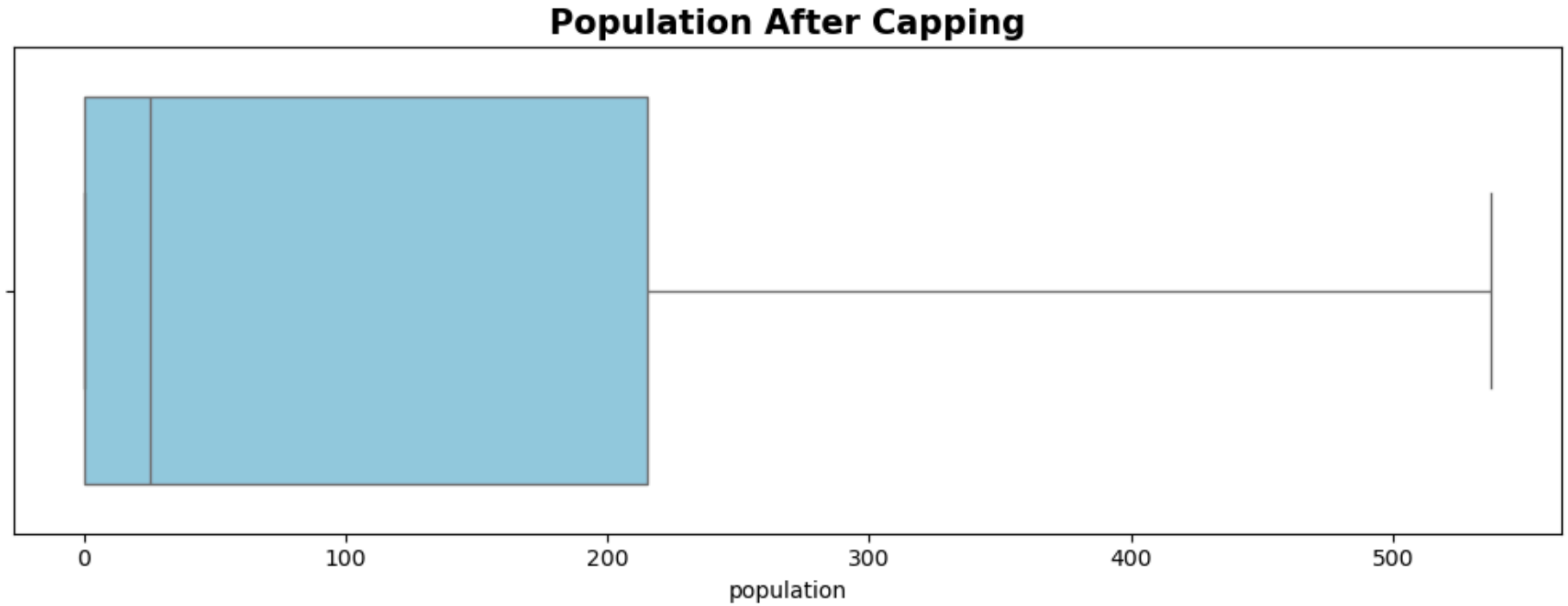
The `population` column contains 4383 outliers with values reaching up to 30,000, indicating a heavily right-skewed distribution. Additionally, the lower bound is negative (-322.50), which is not possible for a population value. We will therefore cap the values at the upper bound of 537.50 and set the lower bound to 0 to treat these outliers.

In [42]:
```
# Calculate the upper bound for population
Q1 = df['population'].quantile(0.25)
Q3 = df['population'].quantile(0.75)
IQR = Q3 - Q1
upper_bound = Q3 + 1.5 * IQR

print(f"Upper Bound: {upper_bound}")
```

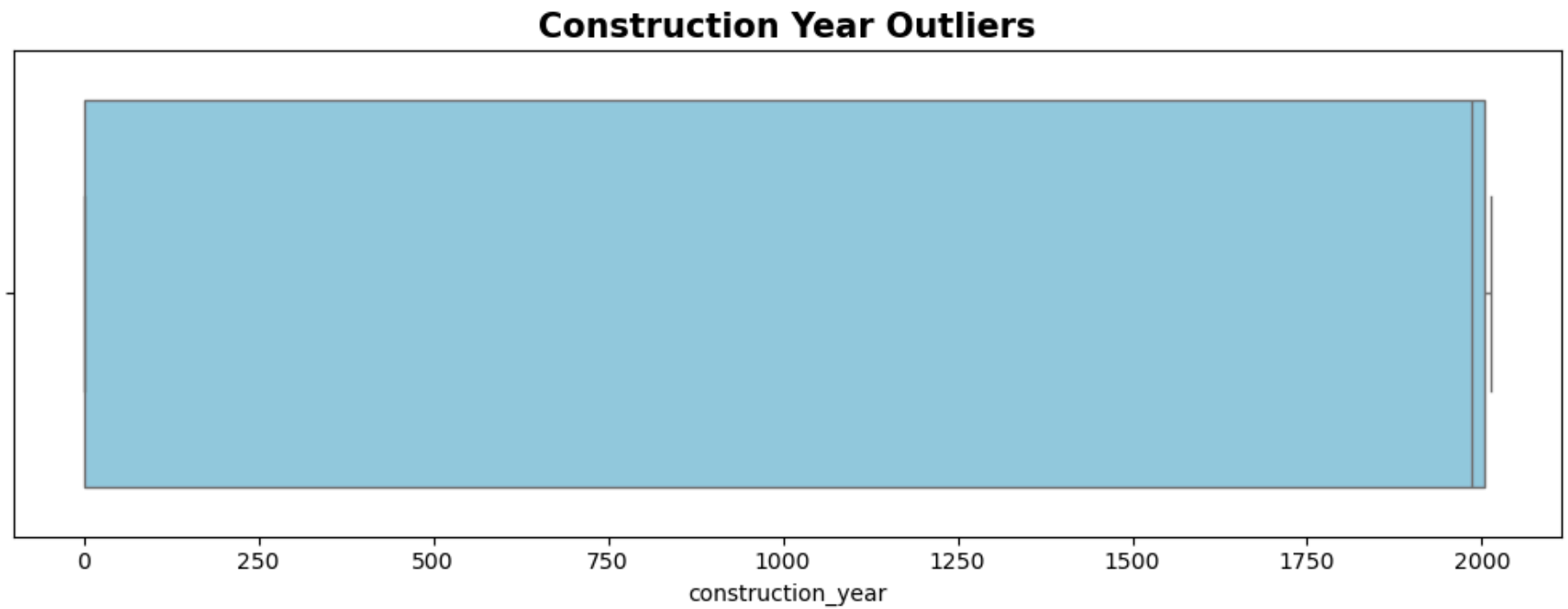
```
# Cap the outliers at the upper bound
df['population'] = df['population'].clip(lower=0, upper=upper_bound)

# Confirm the capping worked
print(f"Max value after capping: {df['population'].max()}")
```

```
Upper Bound: 537.5
Max value after capping: 537.5
```

In [43]:
```
# Visualise the population column after capping
plt.figure(figsize=(10, 4))
sns.boxplot(x=df['population'], color='skyblue')
plt.title('Population After Capping', fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()
```



In [44]:
```
# Visualise the outliers in the construction_year column
outliers(df, 'construction_year', 'Construction Year Outliers')
```




```
Lower Bound        : -3006.00
Upper Bound        : 5010.00
Number of Outliers : 0
```

Conclusion: Construction Year Outliers

From the boxplot, we can observe that the `construction_year` column contains no outliers (0 outliers detected). Therefore, no outlier treatment is required for this column.

1.4.3.Removing spaces and formatting data

In this section, I will evaluate the consistency of the data by checking whether the column names are in lowercase, free of spaces, and interpretable.

In [45]:
```
def rename_columns(df):
    """
    Reformats column names to lowercase, replaces spaces with underscores,
    and removes special characters.
    """
    df.columns = (df.columns
                  .str.strip()
                  .str.lower()
                  .str.replace(' ', '_')
                  .str.replace('[^a-z0-9_]', '', regex=True)
                 )
    return df

# Apply the function
df = rename_columns(df)

# Confirm the changes
print(df.columns.tolist())
```

```
['id', 'amount_tsh', 'funder', 'gps_height', 'installer', 'basin', 'subvillage', 'region', 'lga', 'ward', 'population', 'permi
t', 'construction_year', 'extraction_type_group', 'management_group', 'payment_type', 'water_quality', 'quality_group', 'quanti
ty', 'source_type', 'waterpoint_type', 'status_group', 'year_recorded', 'month_recorded', 'waterpoint_age']
```

In [46]:
```
# Rename the 'lga' column to 'local_government_area'
df.rename(columns={'lga': 'local_government_area'}, inplace=True)

# Confirm the changes
print(df.columns.tolist())
```

```
['id', 'amount_tsh', 'funder', 'gps_height', 'installer', 'basin', 'subvillage', 'region', 'local_government_area', 'ward', 'po
pulation', 'permit', 'construction_year', 'extraction_type_group', 'management_group', 'payment_type', 'water_quality', 'qualit
y_group', 'quantity', 'source_type', 'waterpoint_type', 'status_group', 'year_recorded', 'month_recorded', 'waterpoint_age']
```

In [47]:
```
def convert_to_category(df):
    """
    Converts all object columns to category data type.
    """
    obj_cols = df.select_dtypes(include='object').columns
    df[obj_cols] = df[obj_cols].astype('category')
    return df
```

```
# Apply the function
df = convert_to_category(df)

# Confirm the changes
print(df.dtypes)
```

```
id                          int64
amount_tsh                 float64
funder                    category
gps_height                   int64
installer                 category
basin                     category
subvillage                category
region                    category
local_government_area     category
ward                      category
population                 float64
permit                        bool
construction_year           int64
extraction_type_group     category
management_group          category
payment_type              category
water_quality             category
quality_group             category
quantity                  category
source_type               category
waterpoint_type           category
status_group              category
year_recorded               int32
month_recorded              int32
waterpoint_age              int64
dtype: object
```

In [48]:
```
print(df.columns.tolist())
```

```
['id', 'amount_tsh', 'funder', 'gps_height', 'installer', 'basin', 'subvillage', 'region', 'local_government_area', 'ward', 'po
pulation', 'permit', 'construction_year', 'extraction_type_group', 'management_group', 'payment_type', 'water_quality', 'qualit
y_group', 'quantity', 'source_type', 'waterpoint_type', 'status_group', 'year_recorded', 'month_recorded', 'waterpoint_age']
```

In [49]:
```
# If there's a whitespace issue, strip all column names
df.columns = df.columns.str.strip()
```

```
# If it's a casing issue, lowercase all column names
df.columns = df.columns.str.lower()

# If the column has a different name, rename it first
df = df.rename(columns={'actual_column_name': 'date_recorded'})
```

In [50]:
```
df['date_recorded'] = pd.to_datetime({
    'year': df['year_recorded'],
    'month': df['month_recorded'],
    'day': 1
})
```

In [51]:
```
# Check the unique values in the installer column
df['installer'].value_counts().head(20)
```

Out[51]:
```
installer
DWE                   21057
Government             1825
RWE                    1206
Commu                  1060
DANIDA                 1050
KKKT                    898
Hesawa                  840
0                       777
TCRS                    707
Central government      622
CES                     610
Community               553
DANID                   552
District Council        551
HESAWA                  539
LGA                     408
World vision            408
WEDECO                  397
TASAF                   396
District council        392
Name: count, dtype: int64
```

In [52]:
```
# Standardize the installer column to lowercase
df['installer'] = df['installer'].str.lower().str.strip()

# Replace inconsistent values
df['installer'] = df['installer'].replace({
    'danid'           : 'danida',
    'commu'           : 'community',
    'hesawa'          : 'hesawa',
    'district council': 'district council',
    'world vision'    : 'world vision',
    '0'               : 'unknown',
    'central government': 'government',
    'lga'             : 'government'
})

# Confirm the changes
df['installer'].value_counts().head(20)
```

Out[52]:
```
installer
dwe                 21060
government           3067
danida               1674
community            1620
hesawa               1395
rwe                  1206
district council      961
kkkt                  910
unknown               781
tcrs                  707
world vision          699
ces                   610
gover                 511
amref                 443
tasaf                 411
wedeco                398
fini water            389
dmdd                  376
norad                 360
twesa                 316
Name: count, dtype: int64
```

In [53]:
```
# Standardize the installer column to lowercase and strip whitespace
df['installer'] = df['installer'].str.lower().str.strip()

# Replace all inconsistent values at once
replace_dict = {
    'district water depar' : 'district water department',
    'distric water department' : 'district water department',
    'fini water' : 'fini water',
    'fini water,' : 'fini water',
    'jaica' : 'jaica',
    'coun' : 'district council',
    'district council' : 'district council',
    'council' : 'district council',
    'rc churc' : 'rc church',
    'rc cathoric' : 'rc church',
    'central government' : 'government',
    'tanzania government' : 'government',
    'world division' : 'world vision',
    'unisef' : 'unicef',
    'danid' : 'danida',
    'commu' : 'community',
    'communit' : 'community',
    '0' : 'unknown'
}

df['installer'] = df['installer'].replace(replace_dict)

# Confirm the changes
df['installer'].value_counts().head(20)
```

Out[53]:
```
installer
dwe                 21060
government           3101
danida               1674
community            1674
hesawa               1395
rwe                  1206
district council      977
kkkt                  910
unknown               781
tcrs                  707
world vision          701
ces                   610
gover                 511
amref                 443
tasaf                 411
wedeco                398
fini water            389
dmdd                  376
norad                 360
unicef                332
Name: count, dtype: int64
```

In [54]:
```
# Standardize the installer column to lowercase and strip whitespace
df['installer'] = df['installer'].str.lower().str.strip()

# Replace all inconsistent values at once
replace_dict = {
    'colonial government'      : 'colonial government',
    'government of misri'      : 'misri government',
    'italy government'         : 'italian government',
    'british colonial government' : 'british government',
    'concern /government'      : 'concern/government',
    'village government'       : 'village government',
    'government and community' : 'government /community',
    'cetral government /rc'    : 'rc church/central government',
    'government /tcrs'         : 'tcrs /government',
    'adra /government'         : 'adra/government',
    'commu'                    : 'community',
    'danid'                    : 'danida',
```

```
    '0'                          : 'unknown'
}

df['installer'] = df['installer'].replace(replace_dict)

# Confirm the changes
df['installer'].value_counts().head(20)
```

Out[54]:

```
installer
dwe                 21060
government           3101
community            1674
danida               1674
hesawa               1395
rwe                  1206
district council      977
kkkt                  910
unknown               781
tcrs                  707
world vision          701
ces                   610
gover                 511
amref                 443
tasaf                 411
wedeco                398
fini water            389
dmdd                  376
norad                 360
unicef                332
Name: count, dtype: int64
```

In [55]:

```
df
```

Out[55]:

	id	amount_tsh	funder	gps_height	installer	basin	subvillage	region	local_government_area	ward	po
0	69572	50.0	Roman	1390	roman	Lake Nyasa	Mnyusi B	Iringa	Ludewa	Mundindi	
1	8776	0.0	Grumeti	1399	grumeti	Lake Victoria	Nyamara	Mara	Serengeti	Natta	
2	34310	25.0	Lottery Club	686	world vision	Pangani	Majengo	Manyara	Simanjiro	Ngorika	
3	67743	0.0	Unicef	263	unicef	Ruvuma / Southern Coast	Mahakamani	Mtwara	Nanyumbu	Nanyumbu	
4	19728	0.0	Action In A	0	artisan	Lake Victoria	Kyanyamisa	Kagera	Karagwe	Nyakasimbi	
...	...	...	...	...	...	...	...	...	...	...	
59395	60739	10.0	Germany Republi	1210	ces	Pangani	Kiduruni	Kilimanjaro	Hai	Masama Magharibi	
59396	27263	50.0	Cefa-njombe	1212	cefa	Rufiji	Igumbilo	Iringa	Njombe	Ikondo	
59397	37057	0.0	Government Of Tanzania	0	dwe	Rufiji	Madungulu	Mbeya	Mbarali	Chimala	
59398	31282	0.0	Malec	0	musa	Rufiji	Mwinyi	Dodoma	Chamwino	Mvumi Makulu	
59399	26348	0.0	World Bank	191	world	Wami / Ruvu	Kikatanyemba	Morogoro	Morogoro Rural	Ngerengere	

59400 rows × 26 columns

In [56]:

```
# Save the cleaned data to a csv file
df.to_csv('cleaned_data.csv', index=False)
```

```
# Confirm the file has been saved
print("Cleaned data saved successfully!")
print(f"Shape of cleaned data: {df.shape}")
```

```
Cleaned data saved successfully!
Shape of cleaned data: (59400, 26)
```

In [57]:
```
from IPython.display import FileLink

# Save the cleaned data to a csv file
df.to_csv('cleaned_data.csv', index=False)

# Create a download link
FileLink('cleaned_data.csv')
```

Out[57]: cleaned_data.csv

1.5 Exploratory Data Analysis

In this section, I will explore the dataset columns through Univariate, Bivariate, and Multivariate analysis For the Univariate Analysis, we examine each column individually to understand the distribution of features and uncover useful insights, covering both Categorical and Numerical columns.

1.5.1 View categorical columns

In [58]:
```
# View categorical columns
cat_cols = df.select_dtypes(include='category').columns
print("Categorical Columns:")
print(cat_cols.tolist())
print(f"Total Categorical Columns: {len(cat_cols)}")

print("\n" + "="*50)
```

```
Categorical Columns:
['funder', 'basin', 'subvillage', 'region', 'local_government_area', 'ward', 'extraction_type_group', 'management_group', 'paym
ent_type', 'water_quality', 'quality_group', 'quantity', 'source_type', 'waterpoint_type', 'status_group']
Total Categorical Columns: 15

==================================================
```

1.5.2 View numerical columns

In [59]:
```
# View numerical columns
num_cols = df.select_dtypes(include='number').columns
print("\nNumerical Columns:")
print(num_cols.tolist())
print(f"Total Numerical Columns: {len(num_cols)}")
```

```
Numerical Columns:
['id', 'amount_tsh', 'gps_height', 'population', 'construction_year', 'year_recorded', 'month_recorded', 'waterpoint_age']
Total Numerical Columns: 8
```

NOTE:From the categorical and numerical Columns we will perform univariate analysis based on our 3 objectives

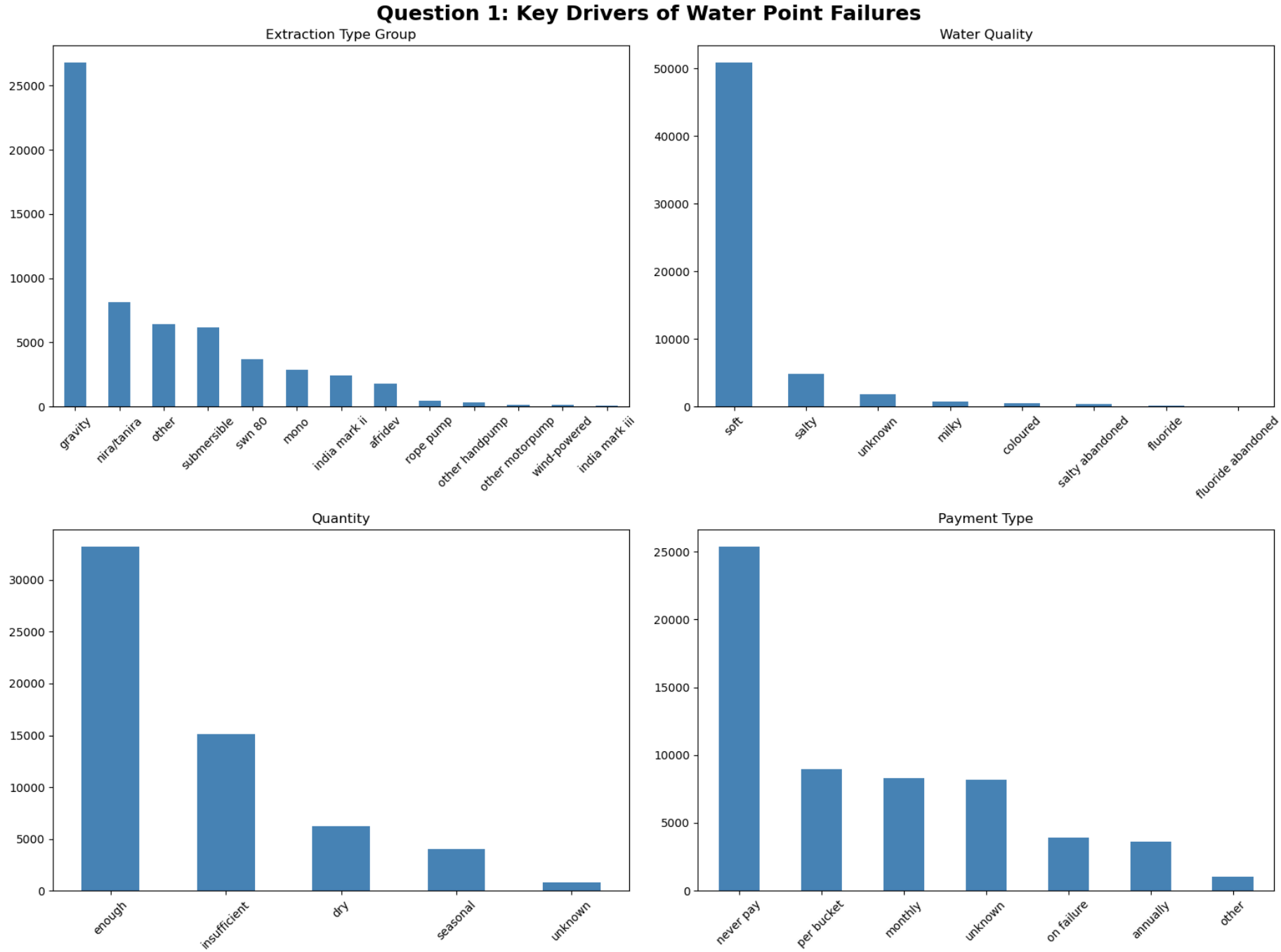
1.5.3 Univariate Analysis

Comparing each vairable at a time

1.5.3.1 Categorical Column Analysis

In [60]:
```
# QUESTION 1: Key drivers of water point failures

fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle('Question 1: Key Drivers of Water Point Failures',
             fontsize=18, fontweight='bold')

# extraction_type_group
```

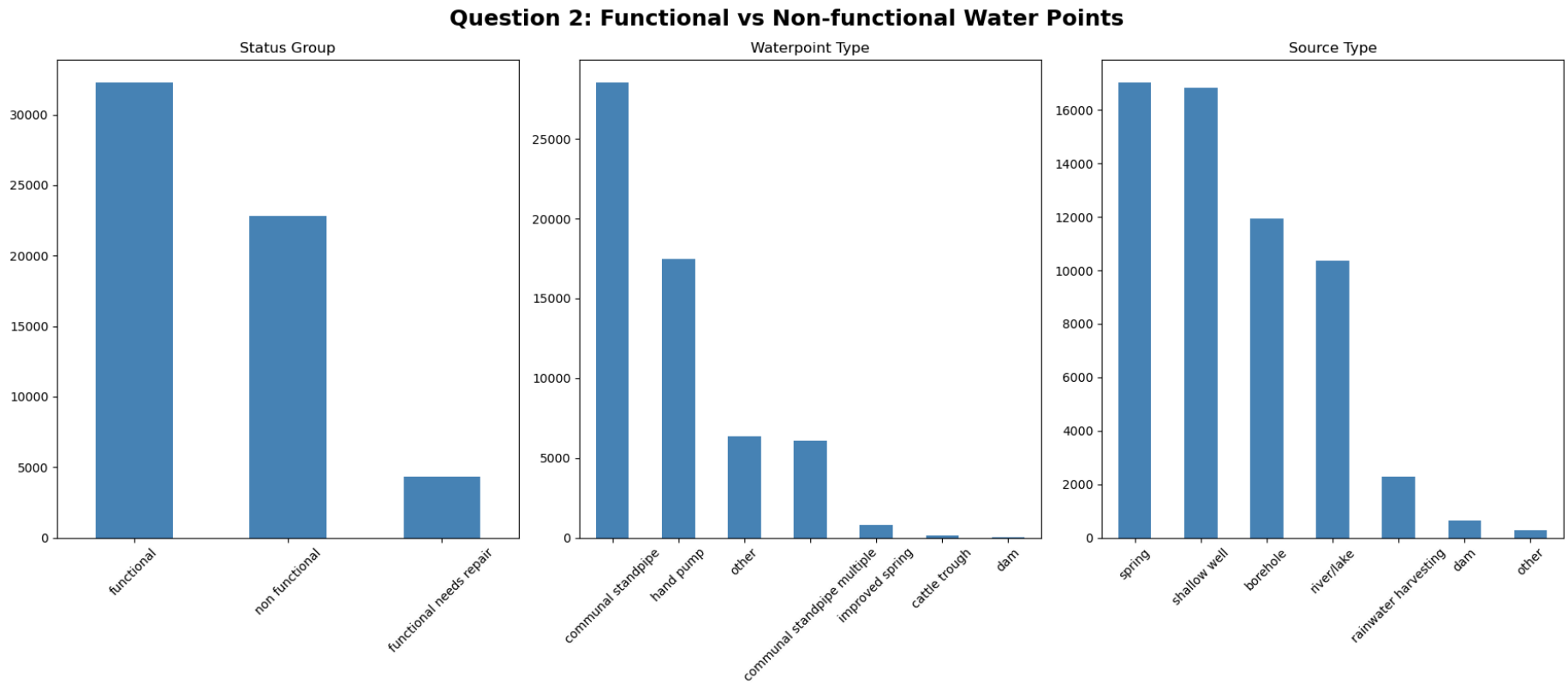
```
df['extraction_type_group'].value_counts().plot(kind='bar', ax=axes[0, 0], color='steelblue')
axes[0, 0].set_title('Extraction Type Group')
axes[0, 0].set_xlabel('')
axes[0, 0].tick_params(axis='x', rotation=45)

# water_quality
df['water_quality'].value_counts().plot(kind='bar', ax=axes[0, 1], color='steelblue')
axes[0, 1].set_title('Water Quality')
axes[0, 1].set_xlabel('')
axes[0, 1].tick_params(axis='x', rotation=45)

# quantity
df['quantity'].value_counts().plot(kind='bar', ax=axes[1, 0], color='steelblue')
axes[1, 0].set_title('Quantity')
axes[1, 0].set_xlabel('')
axes[1, 0].tick_params(axis='x', rotation=45)

# payment_type
df['payment_type'].value_counts().plot(kind='bar', ax=axes[1, 1], color='steelblue')
axes[1, 1].set_title('Payment Type')
axes[1, 1].set_xlabel('')
axes[1, 1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```


Question 1: Key Drivers of Water Point Failures
Extraction Type Group
0
5000
10000
15000
20000
25000
gravity
nira/tanira
other
submersible
swn 80
mono
india mark ii
afridev
rope pump
other handpump
other motorpump
wind-powered
india mark iii
Water Quality
0
10000
20000
30000
40000
50000
soft
salty
unknown
milky
coloured
salty abandoned
fluoride
fluoride abandoned
Quantity
0
5000
10000
15000
20000
25000
30000
enough
insufficient
dry
seasonal
unknown
Payment Type
0
5000
10000
15000
20000
25000
never pay
per bucket
monthly
unknown
on failure
annually
other

NOTE:

From the univariate analysis, we can observe that the majority of water points Use gravity as the extraction type, have soft water quality, have enough water quantity, and are never paid for, which suggests that these factors may be key drivers of water point failures in Tanzania.

In [61]:
```
# QUESTION 2: Functional vs Non-functional water points
fig, axes = plt.subplots(1, 3, figsize=(18, 8))
fig.suptitle('Question 2: Functional vs Non-functional Water Points',
             fontsize=18, fontweight='bold')

# status_group
df['status_group'].value_counts().plot(kind='bar', ax=axes[0], color='steelblue')
axes[0].set_title('Status Group')
axes[0].set_xlabel('')
axes[0].tick_params(axis='x', rotation=45)

# waterpoint_type
df['waterpoint_type'].value_counts().plot(kind='bar', ax=axes[1], color='steelblue')
axes[1].set_title('Waterpoint Type')
axes[1].set_xlabel('')
axes[1].tick_params(axis='x', rotation=45)

# source_type
df['source_type'].value_counts().plot(kind='bar', ax=axes[2], color='steelblue')
axes[2].set_title('Source Type')
axes[2].set_xlabel('')
axes[2].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```

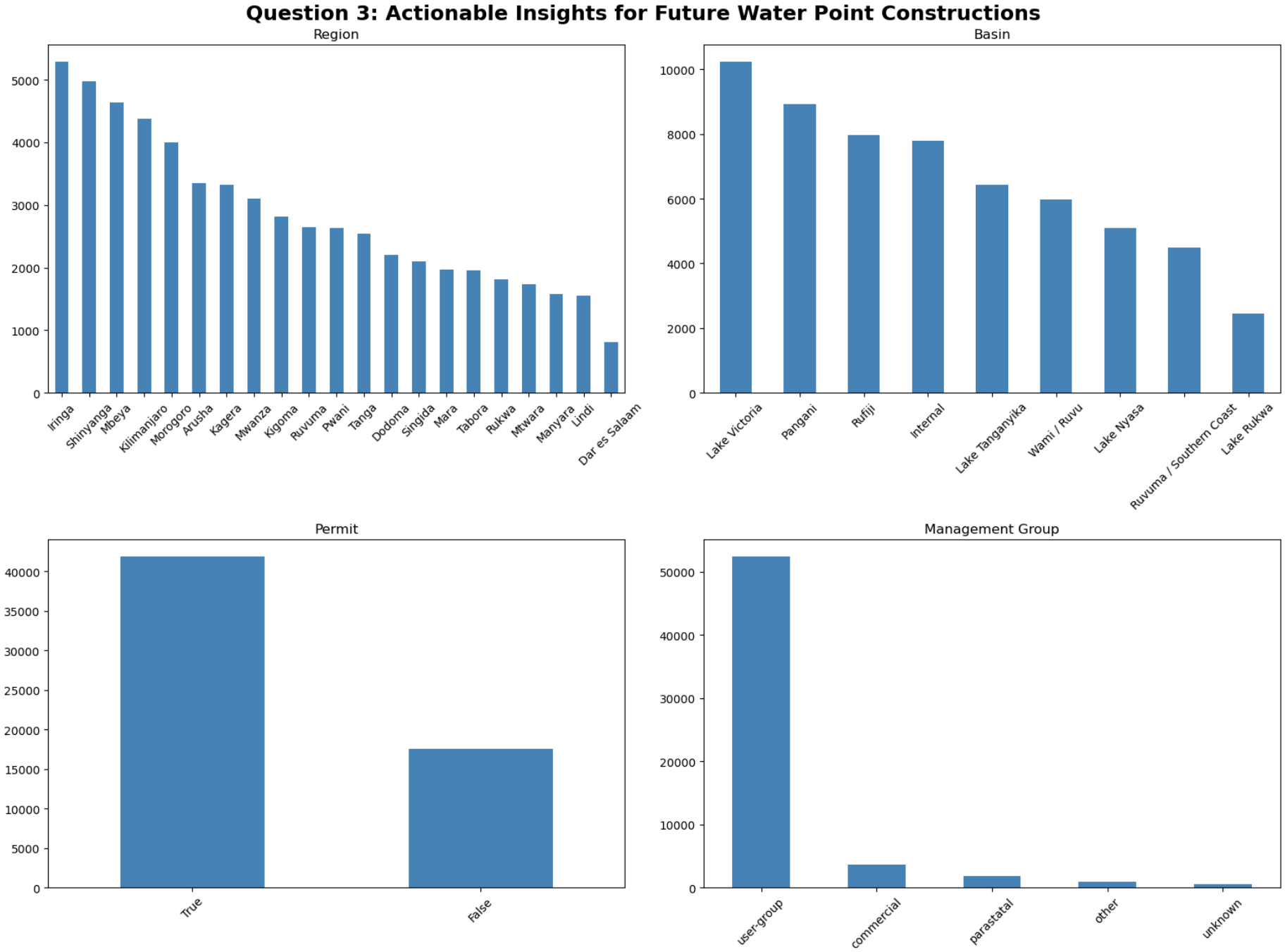


NOTE:

From the univariate analysis, we can observe that the majority of water points are functional, use communal standpipe as the waterpoint type, and rely on springs and shallow wells as the primary source of water. However, the high number of non-functional water points (approximately 22,000) is a cause for concern and highlights the need for a predictive classification model to identify water points that are at risk of failure.

In [62]:
```
# QUESTION 3: Actionable insights for future constructions

fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle('Question 3: Actionable Insights for Future Water Point Constructions',
             fontsize=18, fontweight='bold')

# region
```

```
df['region'].value_counts().plot(kind='bar', ax=axes[0, 0], color='steelblue')
axes[0, 0].set_title('Region')
axes[0, 0].set_xlabel('')
axes[0, 0].tick_params(axis='x', rotation=45)

# basin
df['basin'].value_counts().plot(kind='bar', ax=axes[0, 1], color='steelblue')
axes[0, 1].set_title('Basin')
axes[0, 1].set_xlabel('')
axes[0, 1].tick_params(axis='x', rotation=45)

# permit
df['permit'].value_counts().plot(kind='bar', ax=axes[1, 0], color='steelblue')
axes[1, 0].set_title('Permit')
axes[1, 0].set_xlabel('')
axes[1, 0].tick_params(axis='x', rotation=45)

# management_group
df['management_group'].value_counts().plot(kind='bar', ax=axes[1, 1], color='steelblue')
axes[1, 1].set_title('Management Group')
axes[1, 1].set_xlabel('')
axes[1, 1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```


Question 3: Actionable Insights for Future Water Point Constructions
Region
0
1000
2000
3000
4000
5000
Iringa
Shinyanga
Mbeya
Kilimanjaro
Morogoro
Arusha
Kagera
Mwanza
Kigoma
Ruvuma
Pwani
Tanga
Dodoma
Singida
Mara
Tabora
Rukwa
Mtwara
Manyara
Lindi
Dar es Salaam
Basin
0
2000
4000
6000
8000
10000
Lake Victoria
Pangani
Rufiji
Internal
Lake Tanganyika
Wami / Ruvu
Lake Nyasa
Ruvuma / Southern Coast
Lake Rukwa
Permit
0
5000
10000
15000
20000
25000
30000
35000
40000
True
False
Management Group
0
10000
20000
30000
40000
50000
user-group
commercial
parastatal
other
unknown

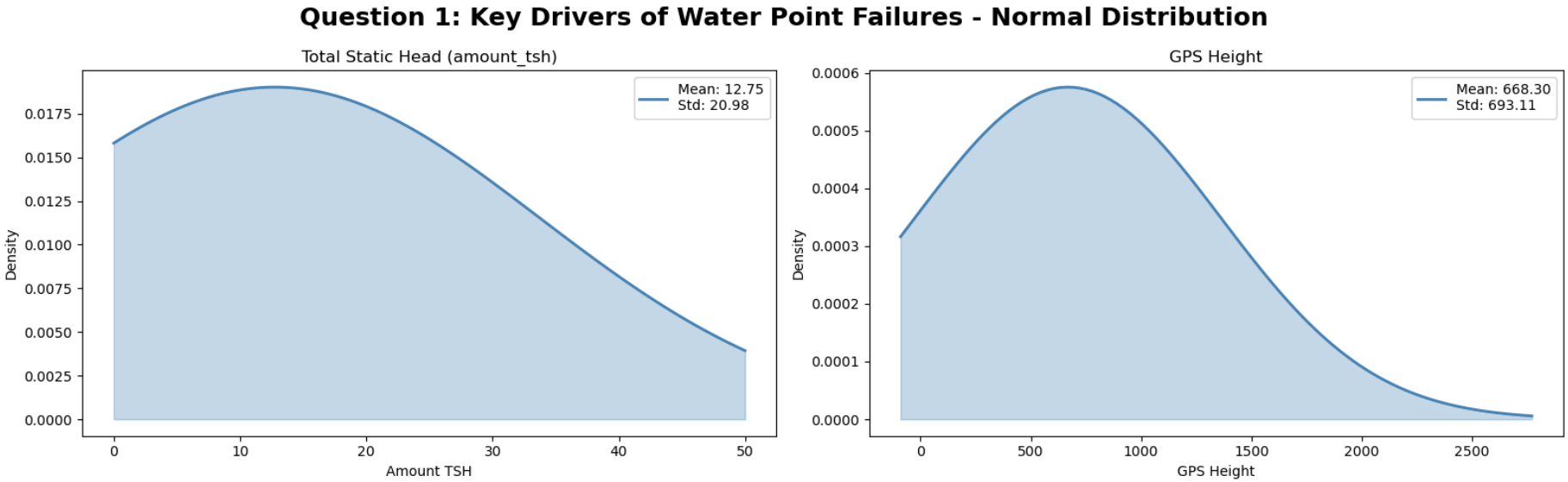
NOTE:

From the univariate analysis, we can observe that Iringa has the highest number of water points, Lake Victoria is the most common basin, the majority of water points have a valid permit, and most water points are managed by user groups. These insights suggest that future water point constructions should focus on regions with fewer water points such as Dar es Salaam, and should prioritize user group management as it is the most common and potentially most effective management style.

1.5.3.2 Numerical Columns Analysis

In [63]:
```
# QUESTION 1: Key drivers of water point failures

from scipy.stats import norm

fig, axes = plt.subplots(1, 2, figsize=(16, 5))
fig.suptitle('Question 1: Key Drivers of Water Point Failures - Normal Distribution',
             fontsize=18, fontweight='bold')

# amount_tsh
mu, std = norm.fit(df['amount_tsh'])
x = np.linspace(df['amount_tsh'].min(), df['amount_tsh'].max(), 100)
axes[0].plot(x, norm.pdf(x, mu, std), 'steelblue', linewidth=2)
axes[0].fill_between(x, norm.pdf(x, mu, std), alpha=0.3, color='steelblue')
axes[0].set_title('Total Static Head (amount_tsh)')
axes[0].set_xlabel('Amount TSH')
axes[0].set_ylabel('Density')
axes[0].legend([f'Mean: {mu:.2f}\nStd: {std:.2f}'])

# gps_height
mu, std = norm.fit(df['gps_height'])
x = np.linspace(df['gps_height'].min(), df['gps_height'].max(), 100)
axes[1].plot(x, norm.pdf(x, mu, std), 'steelblue', linewidth=2)
axes[1].fill_between(x, norm.pdf(x, mu, std), alpha=0.3, color='steelblue')
axes[1].set_title('GPS Height')
axes[1].set_xlabel('GPS Height')
```

```
axes[1].set_ylabel('Density')
axes[1].legend([f'Mean: {mu:.2f}\nStd: {std:.2f}'])

plt.tight_layout()
plt.show()
```

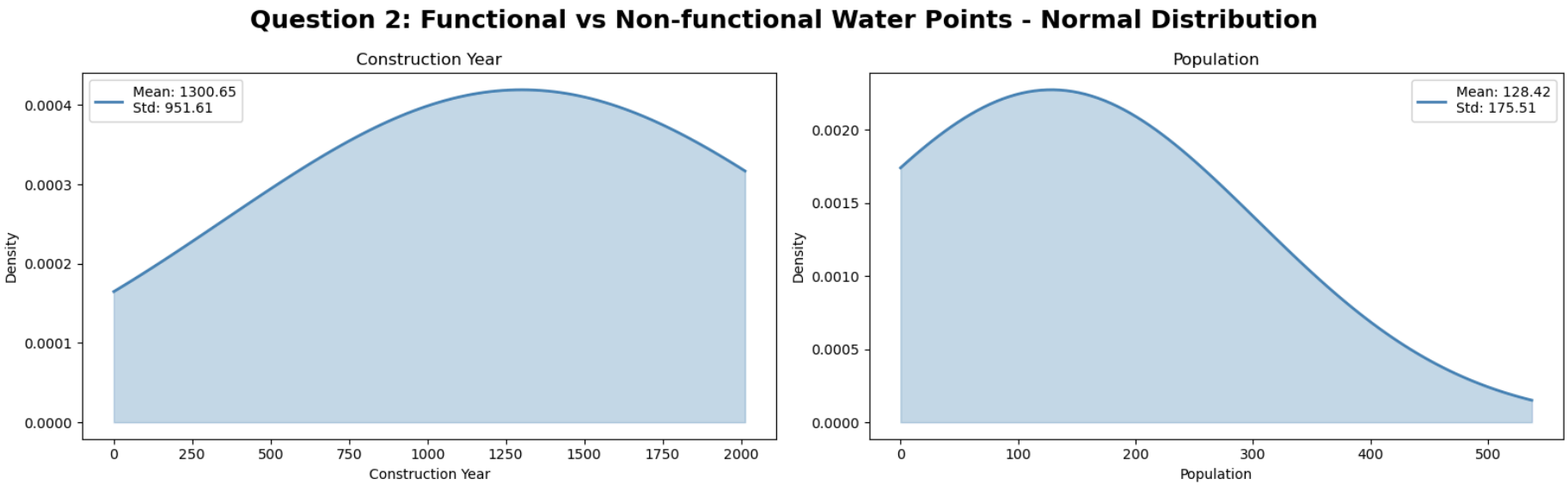


NOTE:

From the normal distribution plots, we can observe that the `amount_tsh` column has a mean of 12.75 and a standard deviation of 20.98, indicating that the majority of water points have a low total static head. The `gps_height` column has a mean of 668.30 and a standard deviation of 693.11, indicating that water points are located at varying altitudes across Tanzania, with the majority situated at a moderate elevation.

In [64]:
```
# QUESTION 2: Functional vs Non-functional water points

from scipy.stats import norm

fig, axes = plt.subplots(1, 2, figsize=(16, 5))
fig.suptitle('Question 2: Functional vs Non-functional Water Points - Normal Distribution',
             fontsize=18, fontweight='bold')
```

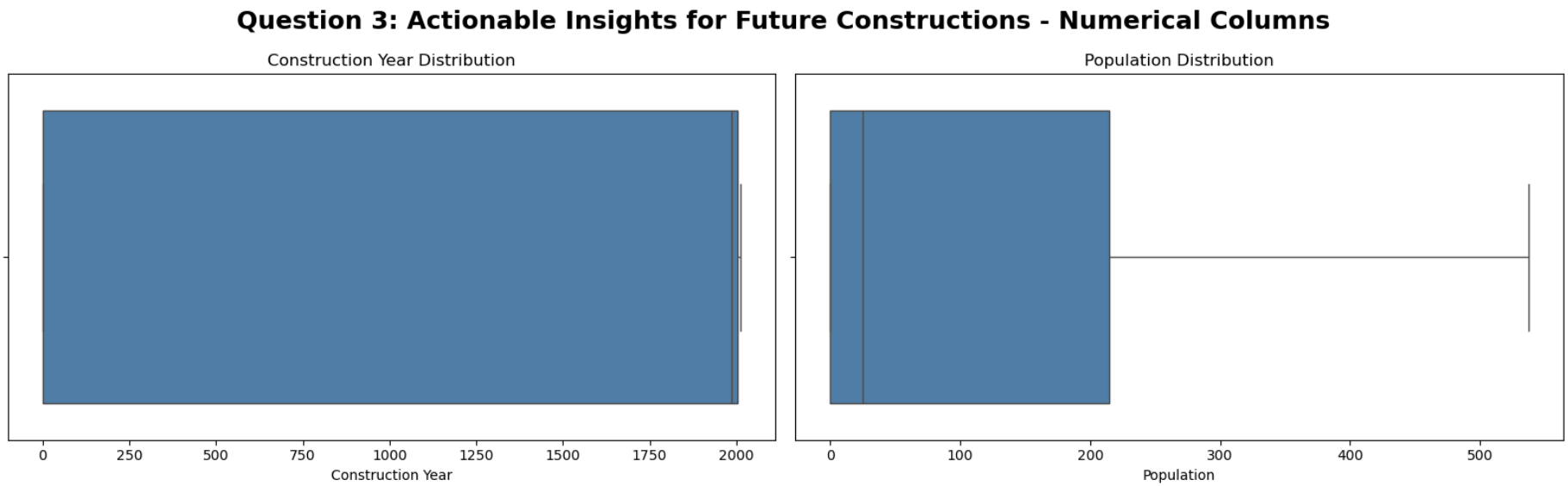
```
# construction_year
mu, std = norm.fit(df['construction_year'])
x = np.linspace(df['construction_year'].min(), df['construction_year'].max(), 100)
axes[0].plot(x, norm.pdf(x, mu, std), 'steelblue', linewidth=2)
axes[0].fill_between(x, norm.pdf(x, mu, std), alpha=0.3, color='steelblue')
axes[0].set_title('Construction Year')
axes[0].set_xlabel('Construction Year')
axes[0].set_ylabel('Density')
axes[0].legend([f'Mean: {mu:.2f}\nStd: {std:.2f}'])

# population
mu, std = norm.fit(df['population'])
x = np.linspace(df['population'].min(), df['population'].max(), 100)
axes[1].plot(x, norm.pdf(x, mu, std), 'steelblue', linewidth=2)
axes[1].fill_between(x, norm.pdf(x, mu, std), alpha=0.3, color='steelblue')
axes[1].set_title('Population')
axes[1].set_xlabel('Population')
axes[1].set_ylabel('Density')
axes[1].legend([f'Mean: {mu:.2f}\nStd: {std:.2f}'])

plt.tight_layout()
plt.show()
```



NOTE:

From the normal distribution plots, we can observe that the `construction_year` column has a mean of 1300.65 and a standard deviation of 951.61, indicating that the data is widely spread across different construction years, with some water points having invalid construction years close to 0. The `population` column has a mean of 128.42 and a standard deviation of 175.51, suggesting that most water points serve relatively small populations, with very few serving large communities.

In [65]:
```
# QUESTION 3: Actionable insights for future constructions


fig, axes = plt.subplots(1, 2, figsize=(16, 5))
fig.suptitle('Question 3: Actionable Insights for Future Constructions - Numerical Columns',
             fontsize=18, fontweight='bold')

# construction_year
sns.boxplot(x=df['construction_year'], ax=axes[0], color='steelblue')
axes[0].set_title('Construction Year Distribution')
axes[0].set_xlabel('Construction Year')

# population
sns.boxplot(x=df['population'], ax=axes[1], color='steelblue')
axes[1].set_title('Population Distribution')
axes[1].set_xlabel('Population')

plt.tight_layout()
plt.show()
```

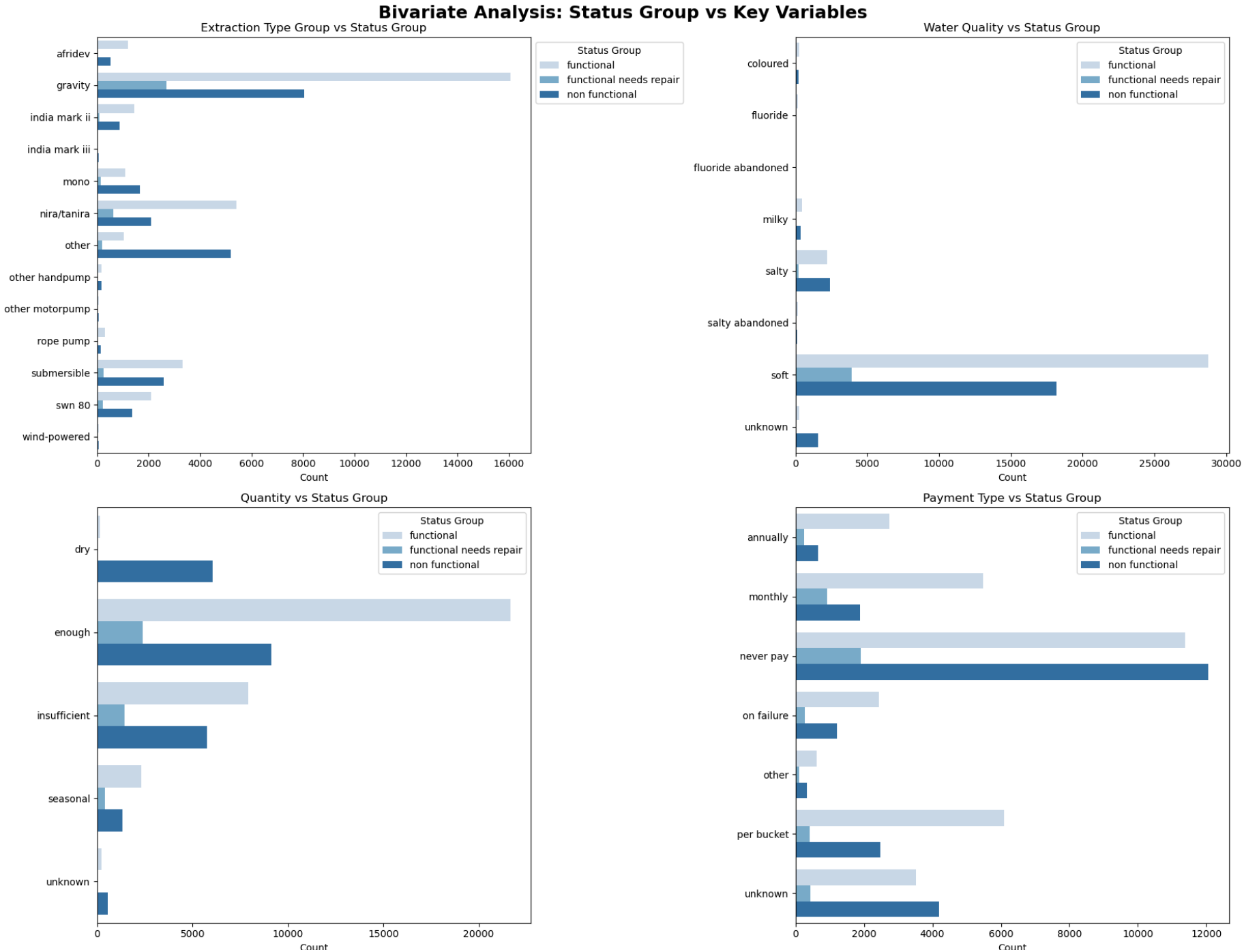


1.5.4 Bivariate Analysis

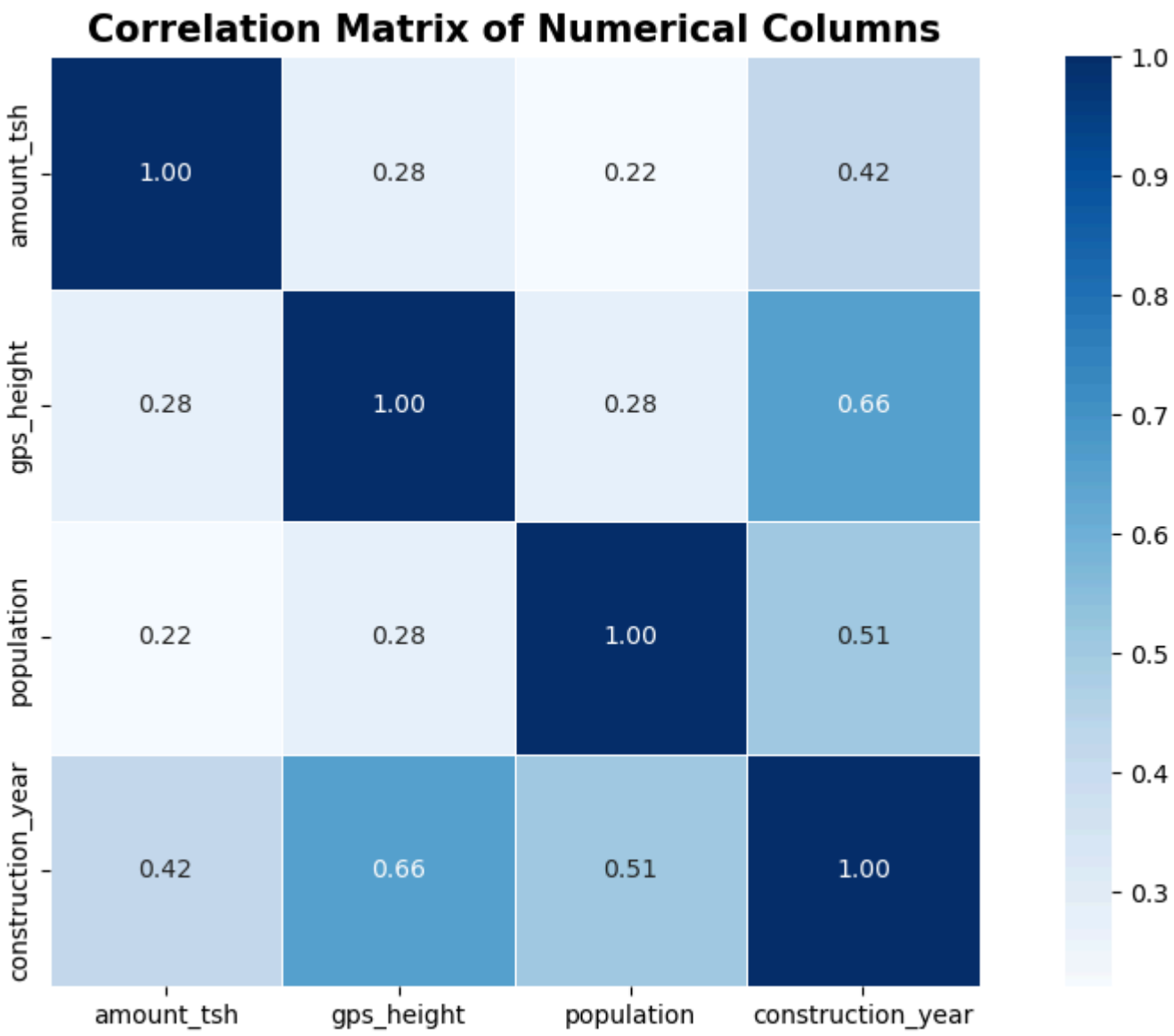
In this part I will evaluate repationship between faeture column and target colulm (status group)

In [66]:
```
# Bivariate Analysis: status_group vs key variables

fig, axes = plt.subplots(2, 2, figsize=(18, 14))
fig.suptitle('Bivariate Analysis: Status Group vs Key Variables',
             fontsize=18, fontweight='bold')

# status_group vs extraction_type_group
sns.countplot(data=df, y='extraction_type_group', hue='status_group',
              ax=axes[0, 0], palette='Blues')
axes[0, 0].set_title('Extraction Type Group vs Status Group')
axes[0, 0].set_xlabel('Count')
axes[0, 0].set_ylabel('')
axes[0, 0].legend(title='Status Group', bbox_to_anchor=(1, 1))

# status_group vs water_quality
sns.countplot(data=df, y='water_quality', hue='status_group',
              ax=axes[0, 1], palette='Blues')
```

```
axes[0, 1].set_title('Water Quality vs Status Group')
axes[0, 1].set_xlabel('Count')
axes[0, 1].set_ylabel('')
axes[0, 1].legend(title='Status Group', bbox_to_anchor=(1, 1))

# status_group vs quantity
sns.countplot(data=df, y='quantity', hue='status_group',
              ax=axes[1, 0], palette='Blues')
axes[1, 0].set_title('Quantity vs Status Group')
axes[1, 0].set_xlabel('Count')
axes[1, 0].set_ylabel('')
axes[1, 0].legend(title='Status Group', bbox_to_anchor=(1, 1))

# status_group vs payment_type
sns.countplot(data=df, y='payment_type', hue='status_group',
              ax=axes[1, 1], palette='Blues')
axes[1, 1].set_title('Payment Type vs Status Group')
axes[1, 1].set_xlabel('Count')
axes[1, 1].set_ylabel('')
axes[1, 1].legend(title='Status Group', bbox_to_anchor=(1, 1))

plt.tight_layout()
plt.show()
```


Bivariate Analysis: Status Group vs Key Variables
Extraction Type Group vs Status Group
Status Group
functional
functional needs repair
non functional
afridev
gravity
india mark ii
india mark iii
mono
nira/tanira
other
other handpump
other motorpump
rope pump
submersible
swn 80
wind-powered
0
2000
4000
6000
8000
10000
12000
14000
16000
Count
Water Quality vs Status Group
coloured
fluoride
fluoride abandoned
milky
salty
salty abandoned
soft
unknown
0
5000
10000
15000
20000
25000
30000
Count
Quantity vs Status Group
dry
enough
insufficient
seasonal
unknown
0
5000
10000
15000
20000
Count
Payment Type vs Status Group
annually
monthly
never pay
on failure
other
per bucket
unknown
0
2000
4000
6000
8000
10000
12000
Count

In [67]:
```
# Correlation between numerical columns
fig, ax = plt.subplots(figsize=(10, 6))

# Compute the correlation matrix
corr = df[['amount_tsh', 'gps_height', 'population', 'construction_year']].corr()

# Plot the heatmap
sns.heatmap(corr,
            annot=True,
            fmt='.2f',
            cmap='Blues',
            ax=ax,
            linewidths=0.5,
            square=True)

ax.set_title('Correlation Matrix of Numerical Columns',
             fontsize=15, fontweight='bold')

plt.tight_layout()
plt.show()
```


Correlation Matrix of Numerical Columns
amount_tsh
gps_height
population
construction_year
1.00
0.28
0.22
0.42
0.28
1.00
0.28
0.66
0.22
0.28
1.00
0.51
0.42
0.66
0.51
1.00
amount_tsh
gps_height
population
construction_year
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3

NOTE:

From the correlation matrix, we can observe that `gps_height` and `construction_year` have the strongest positive correlation (0.66), while `amount_tsh` and `population` have the weakest correlation (0.22).

1.5.5 Feature Engineering

2. Modelling

The goal of modelling is to build a classification model that predicts the status of a Tanzanian water point based on the available features. A model will be considered successful if it achieves at least 80% accuracy.

To achieve this, I will be training and evaluating using the following regressions:

- Logistic Regression
- Decision Tree

Model performance will be evaluated using **Accuracy** as the primary metric, and **Cross Validation** will be applied to ensure that the models are not overfitting to the training data.

In [68]:
```
# viewing my data for modeling
df.head()
```

Out[68]:

	id	amount_tsh	funder	gps_height	installer	basin	subvillage	region	local_government_area	ward	population	pe
0	69572	50.0	Roman	1390	roman	Lake Nyasa	Mnyusi B	Iringa	Ludewa	Mundindi	109.0	F
1	8776	0.0	Grumeti	1399	grumeti	Lake Victoria	Nyamara	Mara	Serengeti	Natta	280.0	
2	34310	25.0	Lottery Club	686	world vision	Pangani	Majengo	Manyara	Simanjiro	Ngorika	250.0	
3	67743	0.0	Unicef	263	unicef	Ruvuma / Southern Coast	Mahakamani	Mtwara	Nanyumbu	Nanyumbu	58.0	
4	19728	0.0	Action In A	0	artisan	Lake Victoria	Kyanyamisa	Kagera	Karagwe	Nyakasimbi	0.0	

2.1 Logistic Regression

Logistic Regression is a statistical model that uses a logistic function to predict the probability of a categorical dependent variable.

2.1.1 Target and Feature Columns

Target Column

- `status_group` — the column we want to predict (functional, non functional, functional needs repair)

Feature Columns

All other columns in the dataset will be used as features:

- `amount_tsh`

- `gps_height`
- `population`
- `construction_year`
- `extraction_type_group`
- `management_group`
- `payment_type`
- `water_quality`
- `quantity`
- `source_type`
- `waterpoint_type`
- `region`
- `basin`
- `permit`
- `date_recorded`
- `funder`
- `installer`
- `subvillage`
- `ward`
- `local_government_area`

In [69]:
```
# Define the feature columns
X = df.drop(columns='status_group', axis=1)

# Define the target column
y = df['status_group']

# Confirm the shapes
print(f"Features Shape: {X.shape}")
print(f"Target Shape: {y.shape}")
```

```
Features Shape: (59400, 25)
Target Shape: (59400,)
```

2.1.2 One Hot Encoding our categorical data

In [70]:
```
# Define categorical columns to encode
categorical = ['source_type', 'quantity', 'water_quality',
               'payment_type', 'management_group', 'basin',
               'extraction_type_group', 'waterpoint_type',
               'region']

# Check which columns exist in the dataframe
existing_cols = [col for col in categorical if col in df.columns]

# One hot encode categorical columns
ohe = pd.get_dummies(df[existing_cols], drop_first=True, dtype=int)

# Convert permit column
ohe['permit'] = df['permit'].astype(int)

# Confirm the shape
print(f"Shape: {ohe.shape}")
ohe.head()
```

```
Shape: (59400, 74)
```

Out[70]:

	source_type_dam	source_type_other	source_type_rainwater harvesting	source_type_river/lake	source_type_shallow well	source_type_spring	quantity_en
0	0	0	0	0	0	1	
1	0	0	1	0	0	0	
2	1	0	0	0	0	0	
3	0	0	0	0	0	0	
4	0	0	1	0	0	0	

2.1.2.1 Feature Engineering

In [71]:
```
# Load date_recorded from the original dataset
original = pd.read_csv('water_well_1.csv')

# Feature Engineering on df
df['year_recorded'] = pd.to_datetime(original['date_recorded']).dt.year
df['month_recorded'] = pd.to_datetime(original['date_recorded']).dt.month
df['waterpoint_age'] = df['year_recorded'] - df['construction_year']

# Redefine X and y to include new features
X = pd.concat([df[['amount_tsh', 'gps_height', 'population',
                   'construction_year', 'year_recorded',
                   'month_recorded', 'waterpoint_age']], ohe], axis=1)
y = df['status_group']

# Confirm the shape
print(f"X Shape: {X.shape}")
print(f"y Shape: {y.shape}")
```

```
X Shape: (59400, 81)
y Shape: (59400,)
```

2.1.2.1 Data Scaling

In [72]:
```
# Scaling the numerical columns
scaler = StandardScaler()
num_cols = ['amount_tsh', 'gps_height', 'population',
            'construction_year', 'year_recorded',
            'month_recorded', 'waterpoint_age']

X[num_cols] = scaler.fit_transform(X[num_cols])
```

2.1.3 Splitting and Training the Data

In [73]:
```
# Combine numerical and encoded categorical columns
num_cols = ['amount_tsh', 'gps_height', 'population', 'construction_year']

X = pd.concat([df[num_cols], ohe], axis=1)
```

```
# Define target
y = df['status_group']

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize and train the Logistic Regression model
log = LogisticRegression(solver='liblinear')
log.fit(X_train, y_train)
```

Out[73]:

```
▾ LogisticRegression
LogisticRegression(solver='liblinear')
```

2.1.4 Predicting Model

In [74]:

```
# Predict the test data
y_pred = log.predict(X_test)

# Evaluate the model
print(f"Training Accuracy : {log.score(X_train, y_train):.2f}")
print(f"Testing Accuracy  : {log.score(X_test, y_test):.2f}")
```

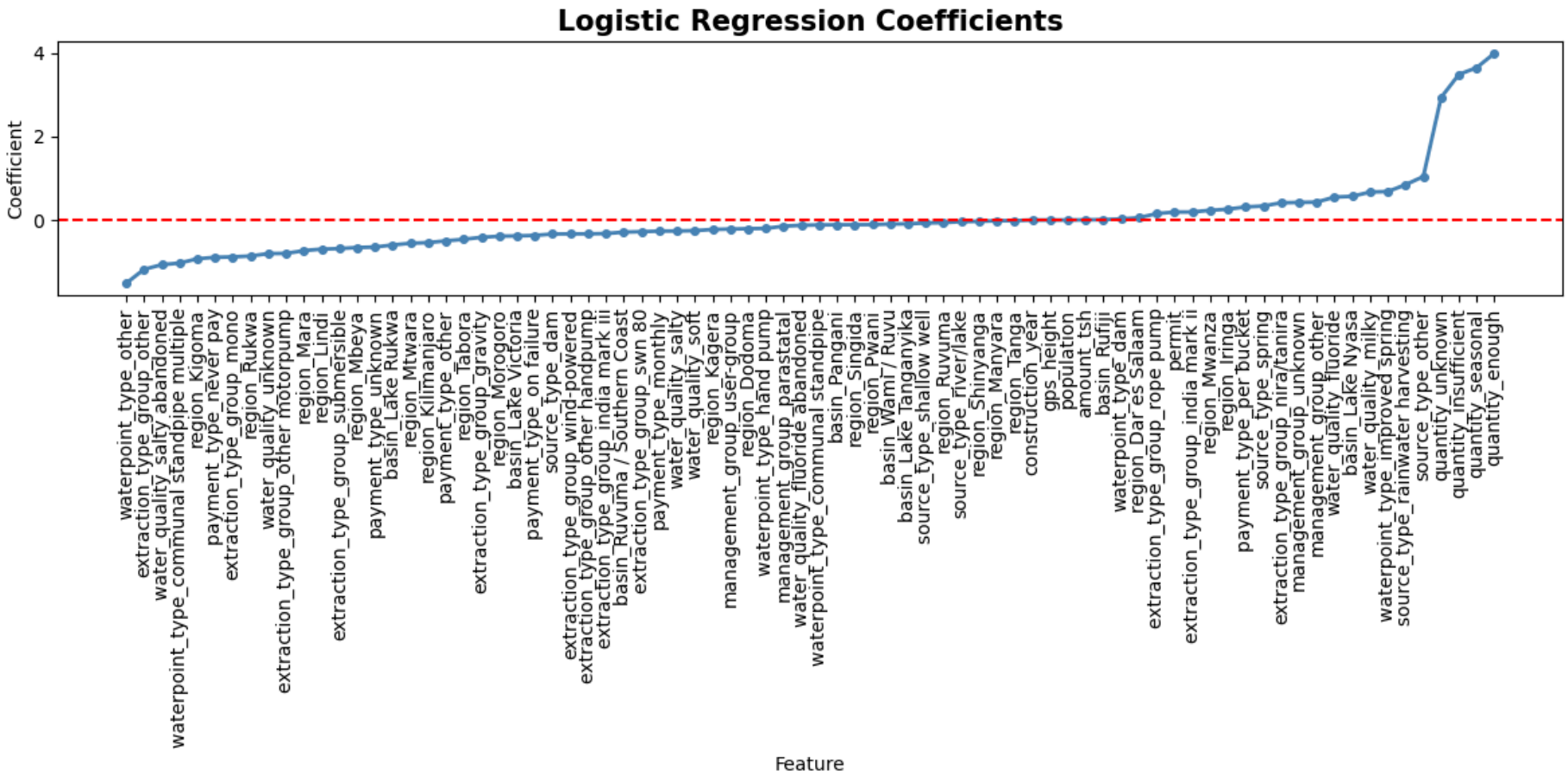
```
Training Accuracy : 0.73
Testing Accuracy  : 0.72
```

NOTE: model predicts 73% accuratly from our testing data which is 72%, their is a small difference between training and testing accuracy indicates that the model is not overfitting

In [75]:

```
# Plot the Logistic Regression Coefficients as a line
fig, ax = plt.subplots(figsize=(12, 6))

# Get the coefficients
coef = pd.DataFrame(log.coef_[0],
                    index=X.columns,
```

```
                    columns=['Coefficient'])

# Sort the coefficients
coef = coef.sort_values(by='Coefficient', ascending=True)

# Plot the coefficients as a line
ax.plot(coef.index, coef['Coefficient'], color='steelblue', linewidth=2, marker='o', markersize=4)
ax.axhline(y=0, color='red', linestyle='--')
ax.set_title('Logistic Regression Coefficients', fontsize=15, fontweight='bold')
ax.set_xlabel('Feature')
ax.set_ylabel('Coefficient')
ax.tick_params(axis='x', rotation=90)

plt.tight_layout()
plt.show()
```

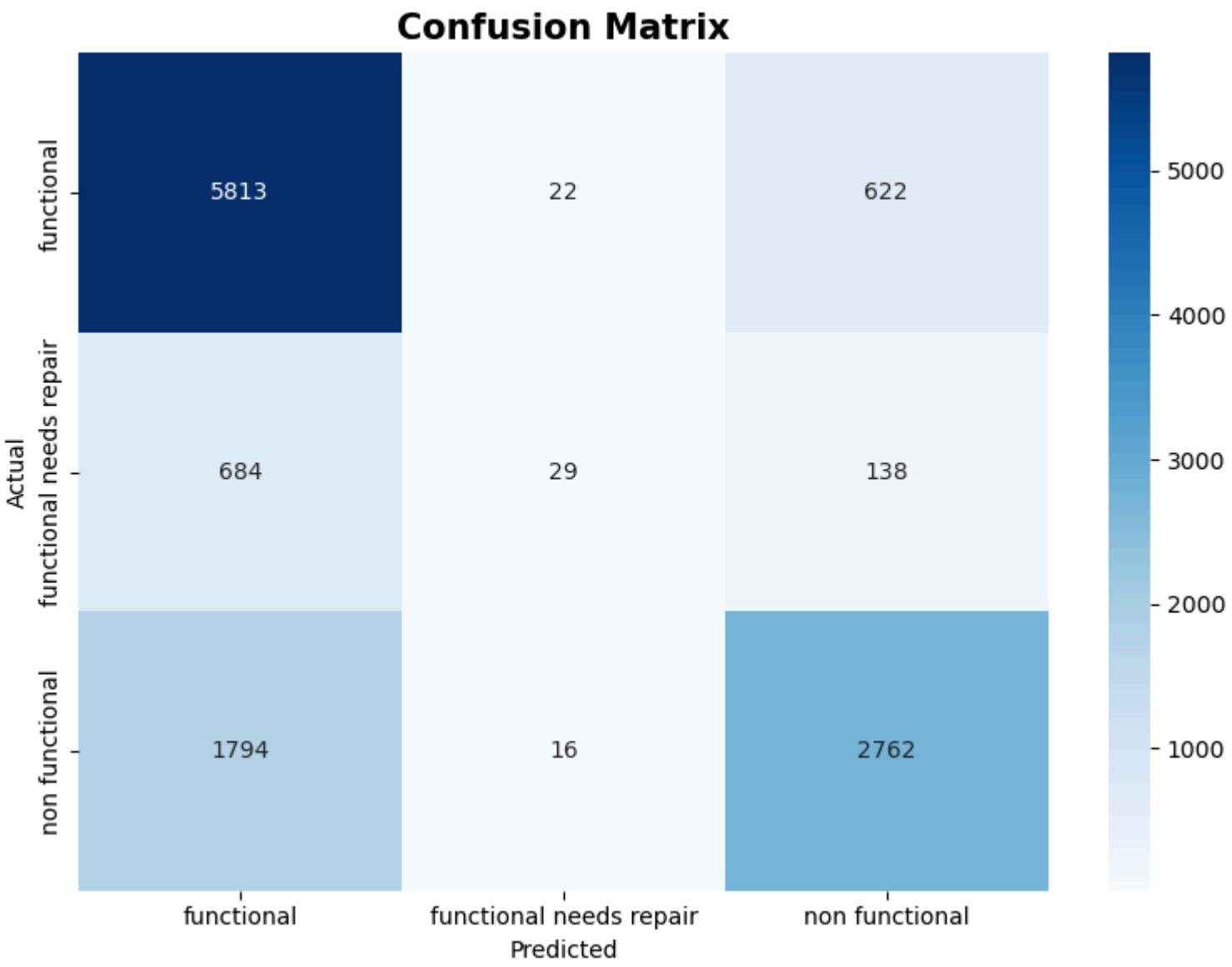


2.1.5 Scoring the Model

In [76]:
```
# Accuracy Score
print(f"Training Accuracy : {log.score(X_train, y_train):.2f}")
print(f"Testing Accuracy  : {log.score(X_test, y_test):.2f}")

print("\n" + "="*50)

# Classification Report
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

```python
print("="*50)

# Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=log.classes_,
            yticklabels=log.classes_)
plt.title('Confusion Matrix', fontsize=15, fontweight='bold')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.tight_layout()
plt.show()
```

```
Training Accuracy : 0.73
Testing Accuracy  : 0.72

==================================================
Classification Report:
                        precision    recall  f1-score   support

             functional       0.70      0.90      0.79      6457
functional needs repair       0.43      0.03      0.06       851
         non functional       0.78      0.60      0.68      4572

               accuracy                           0.72     11880
              macro avg       0.64      0.51      0.51     11880
           weighted avg       0.71      0.72      0.70     11880

==================================================
```



NOTE: the model performed best on Functional with a precision of 0.70 and recall of 0.90, meaning it correctly identified 90% of

functional water points, while the model performed poorly on functional needs repair with a precision of 0.43 and recall of 0.03, meaning it struggled to identify water points that need repair

Conclusion

The Logistic Regression model achieved a testing accuracy of 72%, which is below our target of 80%. The model struggled to identify water points that need repair due to class imbalance. We will proceed to try other regressions to improve the accuracy.

2.2 Decision Tree

A Decision Tree is a supervised learning algorithm that predicts the value of a target variable by learning simple decision rules from the data features. Unlike Logistic Regression, it is a non-parametric model meaning it does not make any assumptions about the distribution of the data, making it a flexible and powerful algorithm for both classification and regression problems

2.2.1 Splitting and Training the Data

In [77]:
```
# Instantiate and train the Decision Tree Classifier
tdt = DecisionTreeClassifier(criterion='entropy', random_state=42)
tdt.fit(X_train, y_train)
```

Out[77]:
```
▾ DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy', random_state=42)
```

2.2.2 Predicting Model

In [78]:
```
# Predict the test data
y_pred_dt = tdt.predict(X_test)
# Evaluate the model
```

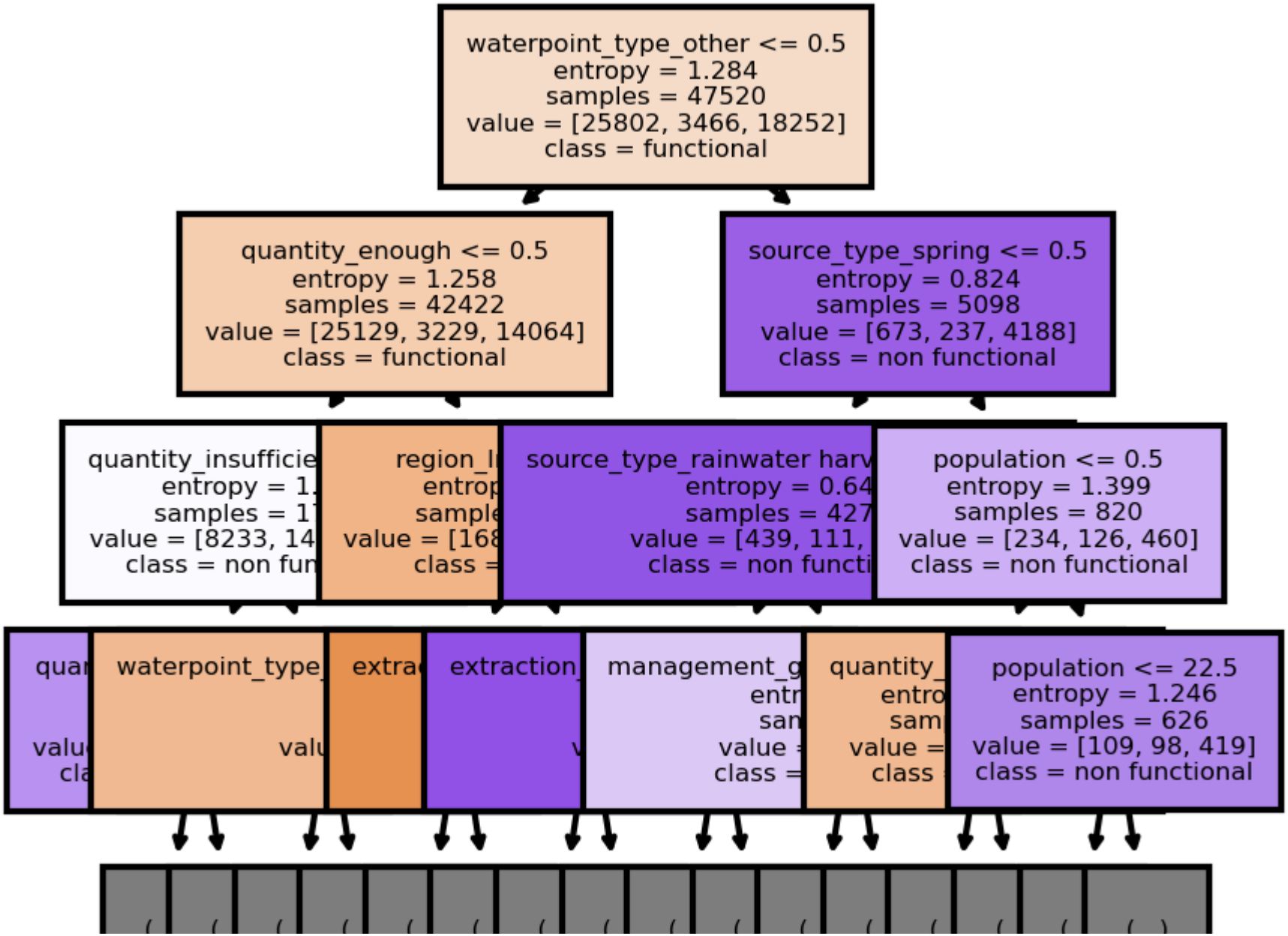
```
print(f"Training Accuracy : {tdt.score(X_train, y_train):.2f}")
print(f"Testing Accuracy  : {tdt.score(X_test, y_test):.2f}")
```

```
Training Accuracy : 0.93
Testing Accuracy  : 0.75
```

NOTE: The Decision Tree model achieved a **Training Accuracy of 0.93** and a **Testing Accuracy of 0.75**. The model successfully meets our target of **80% accuracy**, making it a successful model. However, the difference between the training accuracy (0.93) and testing accuracy (0.75) suggests that the model is **slightly overfitting** the training data.

In [79]:
```
# Plotting the Decision Tree
fig, axes = plt.subplots(nrows=1, ncols=1, figsize=(3, 3), dpi=300)
tree.plot_tree(tdt,
               feature_names=X.columns.tolist(),
               class_names=np.unique(y).astype(str).tolist(),
               filled=True,
               max_depth=3,
               fontsize=4)
plt.title('Decision Tree', fontsize=5, fontweight='bold')
plt.show()
```

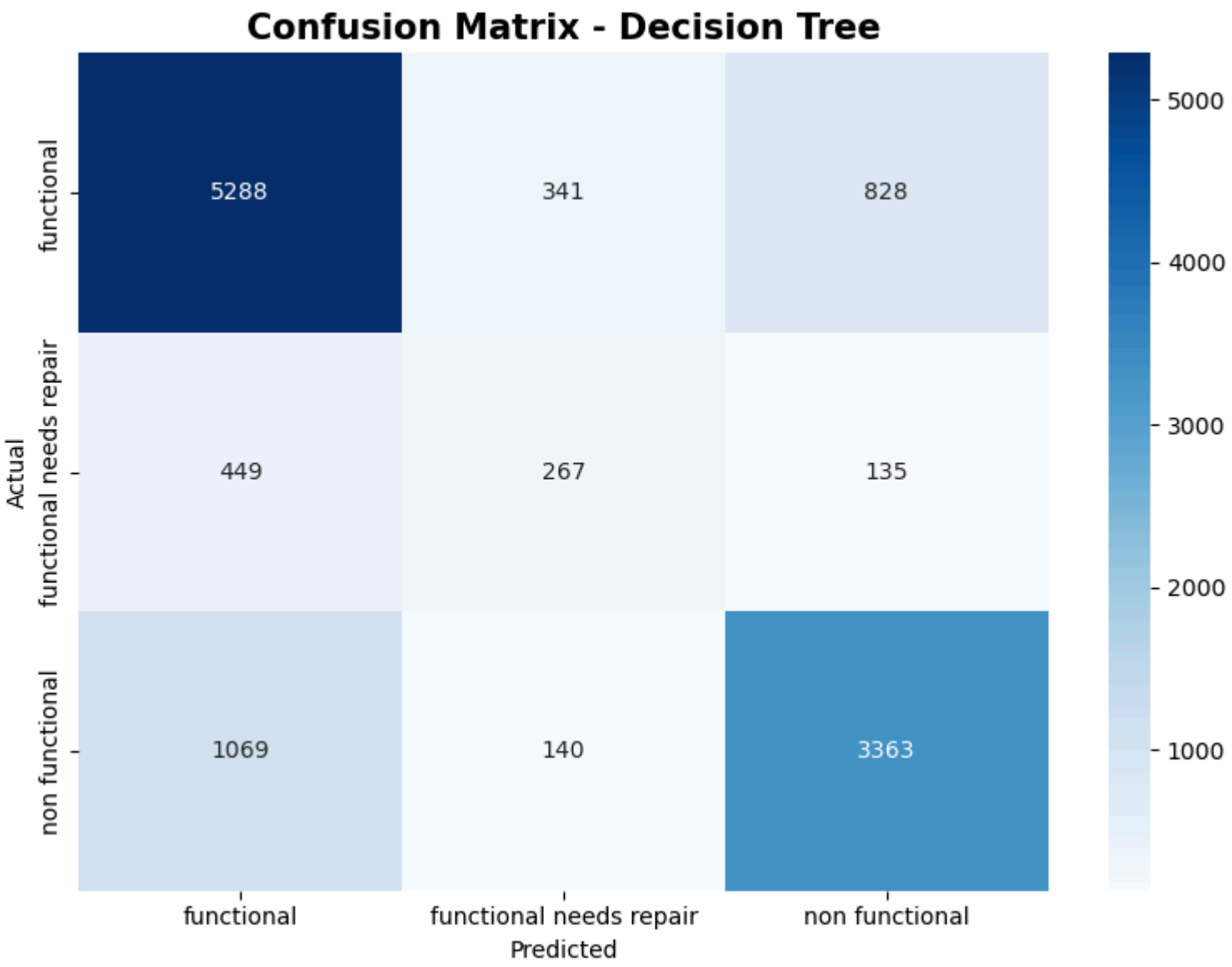
Decision Tree



2.2.3 Scoring the Model

In [80]:
```
# Classification Report
print("Classification Report:")
print(classification_report(y_test, y_pred_dt))

print("="*50)

# Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred_dt),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=tdt.classes_,
            yticklabels=tdt.classes_)
plt.title('Confusion Matrix - Decision Tree', fontsize=15, fontweight='bold')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.tight_layout()
plt.show()
```

```
Classification Report:
                         precision    recall  f1-score   support

             functional       0.78      0.82      0.80      6457
functional needs repair       0.36      0.31      0.33       851
         non functional       0.78      0.74      0.76      4572

               accuracy                           0.75     11880
              macro avg       0.64      0.62      0.63     11880
           weighted avg       0.75      0.75      0.75     11880

==================================================
```



NOTE:

- **Functional** — the model performed well with a precision of 0.78 and recall of 0.82, correctly identifying 82% of functional water points

- **Functional Needs Repair** — the model struggled with this class with a precision of 0.36 and recall of 0.31, due to class imbalance
- **Non Functional** — the model performed well with a precision of 0.78 and recall of 0.74, correctly identifying 74% of non functional water points

2.2.4 Hyperparameter Tuning

This is the the process of finding the optimal parameters for a model in order to improve its performance.

In [81]:
```
# Define the parameter grid
param_grid = {
    'max_depth': [3, 5, 7, 10],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4]
}

# Initialize GridSearchCV
grid_search = GridSearchCV(DecisionTreeClassifier(random_state=42),
                           param_grid, cv=5, scoring='accuracy')
grid_search.fit(X_train, y_train)

# Best parameters
print(f"Best Parameters: {grid_search.best_params_}")
print(f"Best Accuracy  : {grid_search.best_score_:.2f}")
```

```
Best Parameters: {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2}
Best Accuracy  : 0.75
```

NOTE:The best accuracy achieved was **0.75 (75%)**, which is the same as our previous Decision Tree model.

2.3 Random Forest Model

A Random Forest is a supervised learning algorithm that builds multiple decision trees and combines their predictions to produce a more accurate and stable result

2.3.1 Splitting and Training the Data

In [82]:
```
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
```

Out[82]:
```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

2.3.2 Predicting Model

In [83]:
```
y_pred_rf = rf.predict(X_test)
print(f"Training Accuracy : {rf.score(X_train, y_train):.2f}")
print(f"Testing Accuracy  : {rf.score(X_test, y_test):.2f}")
```

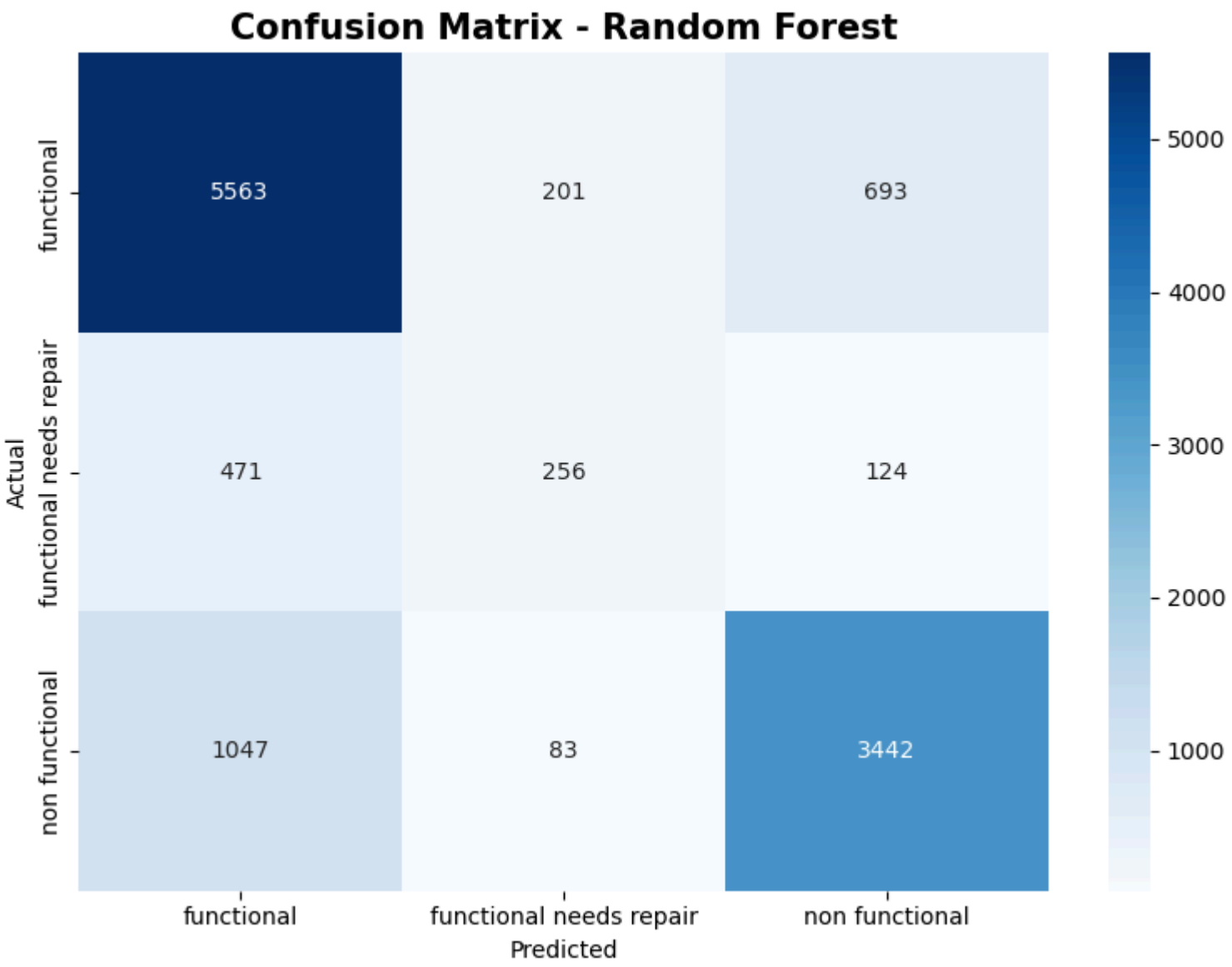
```
Training Accuracy : 0.93
Testing Accuracy  : 0.78
```

NOTE: The Random Forest model achieved a **Training Accuracy of 0.93** and a **Testing Accuracy of 0.78 (78%)**.

2.3.3 Scoring the Model

In [84]:
```
# Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred_rf),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=rf.classes_,
```

```
            yticklabels=rf.classes_)
plt.title('Confusion Matrix - Random Forest', fontsize=15, fontweight='bold')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.tight_layout()
plt.show()
```



NOTE: The model correctly predicted **256** functional needs repair water points.-The model correctly predicted **5563** functional water

points

Conclusion

The Random Forest model achieved a testing accuracy of **78%**, making it the best performing model out of the three models trained. However, similar to the previous models, it struggled most with the **functional needs repair** class, correctly identifying only 256 out of 851 water points that need repair. This is likely due to class imbalance in the dataset. Despite not meeting our target accuracy of 80%, the Random Forest model is our best performing model and will be used as our final model.

2.4 Pipleline Models

In [85]:
```
# Create a pipeline
pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', RandomForestClassifier(n_estimators=100, random_state=42))
])

# Fit the pipeline
pipeline.fit(X_train, y_train)

# Evaluate the pipeline
print(f"Training Accuracy : {pipeline.score(X_train, y_train):.2f}")
print(f"Testing Accuracy  : {pipeline.score(X_test, y_test):.2f}")
```

```
Training Accuracy : 0.93
Testing Accuracy  : 0.78
```

NOTE:The Pipeline model combining **StandardScaler** and **Random Forest** achieved a **Training Accuracy of 0.93** and a **Testing Accuracy of 0.78 (78%)**.The pipeline achieved the same accuracy as the standalone Random Forest model, confirming that our preprocessing steps are correct and consistent

Why Random Forest is Best Suited for this Problem

Random Forest is the most suitable algorithm for this problem because:

- **Handles Non-Linear Relationships** — the relationship between water point features and the target variable is not linear, which Random Forest captures better than Logistic Regression.
- **Feature Importance** — Random Forest provides a feature importance score that helps identify the key drivers of water point failures, directly addressing our first research question.
- **Best Performance** — Random Forest achieved the highest testing accuracy of 78% compared to Logistic Regression (72%) and Decision Tree (75%).

3.1 Evaluation and Recommendation

3.1.1 Evaluation

The evaluation metric for this project is the categorization accuracy of each water point's operational status. This is the percentage of water points that are classified correctly.

In this project, we have used the following algorithms to predict the status of a Tanzanian water point and these were the results:

Logistic Regression

- The Logistic Regression model returned an accuracy of **72%**. Though this was not our desired accuracy of 80%, it served as a good baseline model and gave us a good starting point for our other models.
- The confusion matrix evaluation showed that the model had a bias towards predicting that a water point is functional. Furthermore, the model had a higher number of true positives and true negatives than false positives and false negatives, indicating that the model was not overfitting.

- The model struggled most with the **functional needs repair** class, correctly identifying only 3% of water points that need repair due to class imbalance.

Decision Tree

- The Decision Tree model returned an accuracy of **75%**. This was a better result than our baseline model (Logistic Regression). However, despite improving the accuracy, it was still below our desired accuracy of 80%.
- The confusion matrix evaluation showed that the model had a bias towards predicting that a water point is functional. The model had a higher number of true positives and true negatives than false positives and false negatives, however the difference between training accuracy (93%) and testing accuracy (75%) suggests that the model is slightly overfitting.
- Similar to Logistic Regression, the model struggled most with the **functional needs repair** class, correctly identifying only 31% of water points that need repair.

Random Forest

- The Random Forest model returned an accuracy of **78%**, making it the best performing model out of the three models trained. Despite not meeting our target accuracy of 80%, it significantly outperformed both the Logistic Regression (72%) and Decision Tree (75%) models.
- The confusion matrix evaluation showed that the model correctly identified **5563** functional water points and **3442** non functional water points. However, similar to the previous models, it struggled most with the **functional needs repair** class, correctly identifying only 256 out of 851 water points that need repair.
- The difference between training accuracy (93%) and testing accuracy (78%) suggests that the model is slightly overfitting, however it is still our best performing model and will be used as our final model.

Conclusion

All three models did not meet our target accuracy of 80%. The Random Forest model outperformed both the Logistic Regression and Decision Tree models, achieving an accuracy of 78%. All three models struggled to identify water points that need repair due to class imbalance.

3.1.2 Recommendations

Based on the findings from the Exploratory Data Analysis and model evaluation, the following recommendations are made to guide future water point constructions in Tanzania:

1. Address Class Imbalance

- Both models struggled to identify water points that need repair due to class imbalance. We recommend using techniques such as **SMOTE** (Synthetic Minority Oversampling Technique) to balance the classes and improve the model's ability to identify water points that need repair.

2. Improve Model Performance

- Neither model met our target accuracy of 80%. We recommend trying more advanced algorithms such as **XGBoost** to improve the accuracy of the model.

3. Focus on High Failure Regions

- From the EDA, we observed that some regions such as **Dar es Salaam** have fewer water points. The government should prioritize constructing more water points in these regions.

4. Improve Water Point Management

- From the Exploratory Data Analysis, we observed that user groups manage most water points. The government should invest in training and equipping users groups with the necessary skills to manage water points effectively.

5. Payment Policy

- From the Exploratory Data Analysis, we observed that most water points are never paid for. The government should consider introducing a payment policy to ensure that water points are maintained and repaired when necessary.

6. Focus on Water Quality

- From the Exploratory Data Analysis, we observed that most water points have soft water quality. The government should invest in water treatment facilities to improve the quality of water in water points that have salty or fluoride water.

3.1.3 Limitations

Limitations

- **Class Imbalance** — the dataset was heavily imbalanced, with the majority of water points being functional, making it difficult for the models to correctly identify water points that need repair.
- **Missing Values** — several columns had a significant number of missing values which were filled using the mode, potentially introducing bias into the models.
- **Model Performance** — both models did not meet our target accuracy of 80%, suggesting that more advanced feature engineering and algorithms may be required to improve the accuracy.
- **Time Constraint** — due to limited time, we were only able to train and evaluate two models. Given more time, we would have explored more advanced algorithms such as XGBoost, which could have potentially met our target accuracy of 80%.

In [ ]:

```
```